Learning an Agent-Centered Geometry of Quantum Possibility

Predictive State, Goal-Conditioned Control, and Emergent Spatial Atlases from Measurement Experience Alone

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Abstract

We study a deliberately austere quantum-control problem. An artificial agent may choose a measurement and observe its classical outcome, but it is never given the quantum state, Kraus operators, Born probabilities, Hamiltonian, or tomographic estimates. Its experience is only a sequence of interventions and consequences. The scientific objective is therefore broader than quantum control: what operational representation of an unknown world can be learned from first-person experience, and can many possible aims be organized into a useful geometry?

We compare finite-history tabular Q-learning, goal-conditioned recurrent Q-learning, and a predictive multi-goal gated recurrent unit (GRU). The full agent jointly learns reward value, one-step outcome prediction, a stochastic-shortest-path cost, and goal embeddings regularized by policy similarity. Five environment families vary Hilbert-space dimension, hidden initial state, measurement strength, informational completeness, and outcome alphabet, including a qubit symmetric informationally complete (SIC) measurement and mutually unbiased qutrit bases. Thirty-six agents are trained across six scenarios, three backends, and two seeds.

The predictive geometry agent has the best macro-average success (80.5%), versus 79.1% for finite-history Q-learning and 77.7% for the reward-only GRU, but no backend dominates every environment and neural seed variance is large. Euclidean goal-embedding distance correlates moderately with held-out policy distance (mean Spearman $\rho = 0.54$) and more weakly with independent trajectory distance ($\rho = 0.31$). Learned reachability is directionally informative but not metrically calibrated: blank-history cost correlates $r = 0.47$ with empirical conditional hitting time and has mean absolute error 1.75 interventions. Intervention maps nevertheless reveal a legible directed structure in which outcomes reduce distance to matching goals while often increasing distance to alternatives. Goal geometry is thus a productive diagnostic and planning language, but its metric interpretation requires stronger validation.

We therefore perform a second, inverse-designed study aimed directly at emergent space. A nine-dimensional system supports nine place goals and eight movement instruments. Optimizing diagonal-move success selects $p_d = 0.715$, near $1/\sqrt{2}$, and gives exact 2D stress 0.0365. Three coordinate-free Q-learning agents achieve 100% all-pairs navigation; their learned hitting costs have 1D/2D/3D stresses 0.372/0.071/0.064 and recover the concealed 3×3 arrangement with Procrustes $R^2 = 0.975$. Partial-observability and Manhattan controls identify operational localization and isotropic action cost as necessary ingredients. This constitutes a limited existence proof for stochastic motion in an agent's emergent hodological space and motivates a spatial-base/internal-fiber research program.

Finally, we remove the exact online place report. Four individually ambiguous binary quantum-nondemolition beacons provide a 48-outcome scan history; a GRU predicts a delayed terminal landmark, and an empirically learned transition model supports belief-state planning through blind movements. Across three seeds, delayed-landmark accuracy is 0.964 ± 0.008 and held-out all-pairs navigation is 0.973 ± 0.004, compared with 1.000 ± 0.001 for an online localization oracle. Policy costs have 2D stress 0.075 ± 0.005, correlate 0.948 ± 0.013 with exact stochastic costs, and recover concealed coordinates with Procrustes $R^2 = 0.987 \pm 0.005$. Equal-budget short-memory and place-independent-sensor controls fail. This predictive atlas strengthens the existence result: place can be an anticipated future experience maintained from history rather than a supplied current symbol. Delayed landmark supervision, landmark-anchored surveys, commuting sensors, and expensive fixed scans remain explicit limitations.

Keywords: reinforcement learning; quantum measurement; partial observability; predictive state; goal-conditioned control; reachability; representation learning; QBism; hodological space; multidimensional scaling; emergent space

Contents

1 Introduction

Quantum control is usually described from the standpoint of an external modeler, who writes down a state ρ, a Hamiltonian, and control or measurement operators before optimizing a protocol. Model-free quantum control instead asks what can be learned through interaction, an approach already used for quantum feedback, state preparation, and error correction [19–21].

This work imposes a stricter informational boundary. The agent chooses an intervention a_t and receives a classical outcome o_t. It is not given ρ_t, measurement operators, expectation values, fidelities, or tomography. Its information at time t is

$$h_t = (a_1, o_1, \ldots, a_{t-1}, o_{t-1}). \tag{1}$$

This interface is compatible with QBism's emphasis on an agent's actions, experiences, and expectations [1], although the algorithms do not assume or establish any interpretation of quantum theory. Agent-centered here describes the information boundary.

The central difficulty is partial observability. An external physicist may use ρ_t as a Markov state, but the learner cannot. It needs an internal summary $z_t = F(h_t)$ retaining distinctions relevant to prediction and control. This motivates three objects that must remain distinct:

1. **predictive state**: which histories imply the same future intervention/outcome statistics;
2. **strategic goal geometry**: which goals call for similar policies or trajectories;
3. **directed reachability**: how difficult a goal is from the agent's present history.

Our methodological contribution is a validation framework comparing learned Euclidean goal embeddings with held-out policy behavior, trajectory signatures, empirical hitting times, finite-horizon success curves, and intervention-induced changes in all goal distances.

The paper is written to serve two audiences. Readers primarily interested in reinforcement learning may regard the quantum simulator as an unusual partially observed environment and begin with section 3. Readers primarily interested in quantum foundations may use section 2 to see exactly which quantities are available to the external simulator and which are withheld from the agent. No prior knowledge of reinforcement learning is assumed: we derive the value functions, Bellman recursions, temporal-difference update, recurrent state construction, and evaluation criteria used below. Likewise, the spatial study in section 7 derives the Kraus instruments, exact hitting costs, multidimensional-scaling test, and every reported geometric diagnostic. The successor in section 8 then derives weak-beacon localization, delayed predictive supervision, Bayesian belief filtering, and model-based navigation without an exact online place symbol.

The stronger motivation is hodological: for an agent, goals that require few reliable interventions are near and difficult goals are far. Lewin introduced path-structured "life space" as a formal language for goal-directed behavior [22]. Such a space is generally directed and high-dimensional. Familiar physical space would be a special regime in which a large family of place-like goals admits a common low-stress 2D or 3D embedding and learned action histories become trajectories through it. We treat that as an inverse problem over initial states, Kraus instruments, observation coarse-grainings, and goal repertoires, not as an assumption.

It is useful to state the logical order of the claim at the outset. The hidden simulator defines quantum transitions; interaction produces an empirical control problem; successful control defines all-pairs goal costs; those costs are tested for low-dimensional Euclidean structure; and only then are learned policy histories plotted as spatial trajectories. A visually attractive plot at the last stage cannot substitute for competence or metric validation at the earlier stages.

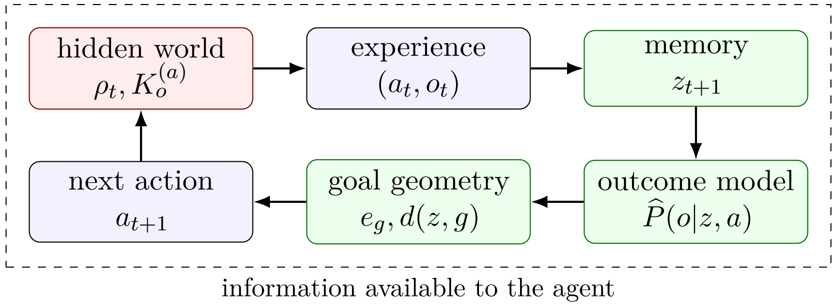


Figure 1: The operational learning loop. Quantum variables in red are private to the simulator. The agent constructs state, prediction, and goal-relative reachability from experienced action/outcome pairs.

Notation and reading guide

Upper-case random variables such as S_t, A_t, O_t distinguish random quantities from realizations s_t, a_t, o_t. A policy is denoted π; the expectation and probability under it are $\mathbb{E}_\pi$ and $\mathbb{P}_\pi$. The hidden quantum state is ρ_t, the observable interaction history is h_t, and a learned or hand-constructed operational state is $z_t = F(h_t)$. Reward value is Q; positive intervention cost is C or d; a goal is g; and an embedding coordinate is x_i. The word *state* is consequently qualified whenever ambiguity matters: quantum state, Markov state, recurrent state, or goal progress state. Distances between goals are symmetric only when explicitly symmetrized. This distinction is central because goal reachability is generally directed.

2 Hidden quantum dynamics

2.1 Quantum instruments and backaction

A finite-dimensional Hilbert space $\mathcal{H} \simeq \mathbb{C}^d$ is the external modeler's state space. A pure state is a unit ray represented by $|\psi\rangle$; uncertainty or classical mixing is represented by a density operator ρ satisfying

$$\rho = \rho^\dagger, \qquad \rho \succeq 0, \qquad \operatorname{Tr}\rho = 1. \tag{2}$$

The notation $\rho \succeq 0$ means $\langle\phi|\rho|\phi\rangle \geq 0$ for every $|\phi\rangle$. A pure state has $\rho = |\psi\rangle\langle\psi|$ and $\operatorname{Tr}\rho^2 = 1$; a genuinely mixed state has $\operatorname{Tr}\rho^2 < 1$. These facts are used by the simulator and its tests, but never supplied as features to the learner.

A measurement action is more completely described by a *quantum instrument* than by a probability distribution. For action a and observed outcome o, let $\mathcal{I}_o^{(a)}$ be a completely positive, trace-nonincreasing map. A Kraus representation is

$$\mathcal{I}_o^{(a)}(\rho) = \sum_\mu K_{o\mu}^{(a)} \rho K_{o\mu}^{(a)\dagger}. \tag{3}$$

The index μ labels physical branches that need not be distinguished by the classical apparatus. Trace preservation after outcomes are forgotten requires [2, 3]

$$\sum_{o,\mu} K_{o\mu}^{(a)\dagger} K_{o\mu}^{(a)} = I. \tag{4}$$

The positive operators $E_o^{(a)} = \sum_\mu K_{o\mu}^{(a)\dagger} K_{o\mu}^{(a)}$ form a POVM, because $E_o^{(a)} \succeq 0$ and $\sum_o E_o^{(a)} = I$. They determine outcome probabilities but not, by themselves, the post-measurement state. The latter depends on the full instrument. The hidden simulator samples and updates by

$$p(o|\rho, a) = \mathrm{Tr}\Big[\mathcal{I}_o^{(a)}(\rho)\Big] = \mathrm{Tr}(E_o^{(a)}\rho), \tag{5}$$

$$\rho' = \frac{\mathcal{I}_o^{(a)}(\rho)}{p(o|\rho, a)}. \tag{6}$$

The normalization follows immediately because $\mathrm{Tr}\,\rho' = \mathrm{Tr}\,\mathcal{I}_o^{(a)}(\rho)/p(o|\rho, a) = 1$. Completeness gives $\sum_o p(o|\rho, a) = \mathrm{Tr}\,\rho = 1$. These elementary checks matter computationally: failure of either condition would turn apparent learning behavior into a simulator artifact.

Measurement therefore does two logically separate things. It produces an experience o, which may inform the agent, and it changes the hidden state, which changes future outcome statistics. The second effect is measurement backaction. If action a makes goal g easy but destroys the preparation needed for goal g', then going from g' toward g need not have the same cost as returning. This is the operational origin of directed reachability in section 4. The learner never evaluates equations (3), (5) and (6); it receives only the action label it chose and the sampled outcome label.

2.2 Projective, unsharp, and informationally complete measurements

For a projective basis $\{|b_o\rangle\}$, one may take $K_o = |b_o\rangle\langle b_o|$. An immediate repetition returns the same outcome with certainty, whereas a measurement in an incompatible basis can erase that certainty. The qubit worlds use Pauli Z, X, and Y bases. For example, $|0\rangle, |1\rangle$ are Z eigenstates and $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$ are X eigenstates, so $|\langle 0|+\rangle|^2 = 1/2$. An X measurement after a known Z outcome is therefore uniformly random and prepares a new X eigenstate. Their noncommutativity creates incompatible preparation strategies and makes action order relevant.

The unsharp binary instruments use axis projectors P_+, P_- and strength $q \in (1/2, 1)$:

$$K_0 = \sqrt{q}\,P_+ + \sqrt{1-q}\,P_-, \qquad K_1 = \sqrt{1-q}\,P_+ + \sqrt{q}\,P_-. \tag{7}$$

Because $P_+P_- = 0$ and $P_+ + P_- = I$, $K_0^\dagger K_0 + K_1^\dagger K_1 = I$. At $q = 0.5$, both effects equal $I/2$: the outcome is uninformative and the state change is correspondingly weak. As $q \to 1$, the operation approaches a projective measurement. We use $q = 0.8$ to create repeated, noisy evidence rather than one-shot certainty.

A SIC-POVM in dimension d contains d^2 equiangular rank-one effects and is informationally complete [4]. For the qubit tetrahedral SIC,

$$E_i = \frac{I + \boldsymbol{r}_i \cdot \boldsymbol{\sigma}}{4}, \qquad \sum_{i=0}^{3} E_i = I, \tag{8}$$

where $\boldsymbol{r}_i$ are tetrahedral Bloch vectors; we use the positive square-root instrument $K_i = \sqrt{E_i}$. This tests mixed outcome alphabets: projective actions have two outcomes while the SIC has four. Invalid logits are masked. Informational completeness means that the probabilities $p_i = \mathrm{Tr}(E_i\rho)$ determine ρ uniquely for an external observer with enough identically prepared samples. It does *not* mean that a single outcome reveals the state, nor does it remove backaction. In the present online problem the learner receives one outcome and the same system continues from the resulting conditional state.

Two bases $\{|e_j\rangle\}$ and $\{|f_k\rangle\}$ are mutually unbiased when $|\langle e_j|f_k\rangle|^2 = 1/d$ for all j, k. Certainty in one becomes a uniform distribution in the other. MUBs formalize complementarity and support tomography [5]. The qutrit world uses three projective MUBs.

Table 1: World/state scenarios. The hidden state is reset each episode.

Environment	d	Initial state	Measurement actions	Goals
qubit-zx-weak	2	$\|1\rangle$	projective Z, X; weak Z	7
qubit-zx-weak	2	$\|+\rangle$	projective Z, X; weak Z	7
qubit-pauli	2	$\|+i\rangle$	projective Z, X, Y	7
qubit-unsharp	2	$I/2$	weak Z, X, Y	6
qubit-pauli-sic	2	$I/2$	projective Z, X, Y; tetrahedral SIC	9
qutrit-mub	3	$\|2\rangle$	three projective MUBs	8

2.3 From hidden quantum state to an observable control process

Combining equations (5) and (6) over time yields a classical distribution over complete interaction histories. If the initial state is ρ_0 and a fixed action sequence is $a_{1:T}$, then the probability of outcomes $o_{1:T}$ is

$$\mathbb{P}(o_{1:T} \mid a_{1:T}, \rho_0) = \mathrm{Tr}\Big(M_T \rho_0 M_T^\dagger\Big), \qquad M_T = K_{o_T}^{(a_T)} \cdots K_{o_1}^{(a_1)}, \tag{9}$$

for one Kraus operator per observed outcome; with coarse-grained branches one sums the corresponding terms. A policy makes the actions themselves depend on past outcomes. The result is an adaptive experiment: by selecting a_t as a function of h_t, the agent changes both what it can learn and what future it creates.

Two hidden density operators are operationally indistinguishable to this agent when they induce the same distribution for every experiment constructible from the available action repertoire. Thus the physically convenient coordinate ρ may contain distinctions irrelevant to the agent, while a short history may omit distinctions essential for future control. The representation problem is to find the quotient between these extremes: enough memory to predict and act, without importing the simulator's private state description.

2.4 Environment catalog

Operational state is relative to the available interventions. With an informationally incomplete repertoire, physically different density operators may imply identical statistics for everything an agent can do. Adding a SIC action or moving from qubit to qutrit may require a richer predictive representation. This motivates varying worlds rather than only random seeds.

3 Reinforcement learning from first principles

3.1 The agent–environment loop

Reinforcement learning (RL) studies how an agent can improve sequential decisions from experience rather than from a supplied transition model [6]. At time t the agent has some information state S_t, draws an action A_t from a policy, the environment produces reward R_{t+1} and a new information state S_{t+1}, and the cycle repeats. An episode is a random trajectory

$$\Xi = (S_0, A_0, R_1, S_1, A_1, R_2, \ldots, S_T). \tag{10}$$

The essential difficulty is delayed consequence. A measurement may have small or negative immediate reward yet prepare a state from which the desired outcome becomes likely several interventions later. Assigning credit only to the final action would miss the preparation strategy.

A policy $\pi(a|s)$ is a conditional probability distribution over actions: $\pi(a|s) \geq 0$ and $\sum_a \pi(a|s) = 1$. A deterministic policy is the special case in which one action has probability one. During training we distinguish the *behavior policy*, which generates data and deliberately explores, from the *target policy*, whose value is being estimated. At evaluation time exploration is disabled so that performance measures the learned decision rule rather than random experimentation.

3.2 Markov decision processes

A finite Markov decision process (MDP) is a tuple

$$(\mathcal{S}, \mathcal{A}, p, r, \gamma), \tag{11}$$

where $\mathcal{S}$ and $\mathcal{A}$ are state and action sets,

$$p(s', r \mid s, a) = \mathbb{P}(S_{t+1} = s', R_{t+1} = r \mid S_t = s, A_t = a) \tag{12}$$

is the time-homogeneous transition/reward kernel, and $\gamma \in [0, 1]$ is a discount factor. The Markov property asserts that once S_t and A_t are known, earlier history supplies no further information about the next transition. It is a sufficiency claim, not a claim that the state is directly visible or ontologically fundamental.

The discounted return from time t is

$$G_t = \sum_{k=0}^{T-t-1} \gamma^k R_{t+k+1} = R_{t+1} + \gamma G_{t+1}. \tag{13}$$

Discounting makes near rewards count more strongly and ensures a finite sum in continuing problems with bounded rewards. In finite episodic problems it also expresses a preference for earlier completion. It should not be confused with the undiscounted intervention count used later for hodological distance.

For a fixed policy, the state and action values are

$$V^\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s], \tag{14}$$

$$Q^\pi(s, a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]. \tag{15}$$

Conditioning on the first transition and substituting equation (13) gives the Bellman expectation equations:

$$V^\pi(s) = \sum_a \pi(a|s) \sum_{s',r} p(s', r|s, a)[r + \gamma V^\pi(s')], \tag{16}$$

$$Q^\pi(s, a) = \sum_{s',r} p(s', r|s, a) \left[r + \gamma \sum_{a'} \pi(a'|s') Q^\pi(s', a') \right]. \tag{17}$$

These are consistency equations: the value of the whole future equals the one-step reward plus the value of the remaining future.

The optimal values $V^*(s) = \max_\pi V^\pi(s)$ and $Q^*(s, a) = \max_\pi Q^\pi(s, a)$ satisfy Bellman's principle of optimality [7]:

$$V^*(s) = \max_a \sum_{s',r} p(s', r|s, a)[r + \gamma V^*(s')], \tag{18}$$

$$Q^*(s, a) = \sum_{s',r} p(s', r|s, a)[r + \gamma \max_{a'} Q^*(s', a')]. \tag{19}$$

Once Q^* is known, any maximizing action is optimal. Dynamic programming would solve these equations using a known p. Model-free RL instead estimates their solution from sampled transitions.

3.3 From Monte Carlo returns to temporal differences

One could wait until an episode ends and use the observed G_t as a noisy target for $Q(S_t, A_t)$. This Monte Carlo estimator is unbiased for the value of the behavior policy, but it has high variance and delays learning. A temporal-difference (TD) method instead *bootstraps*: it uses a current estimate of the remaining value before the episode ends. For Q-learning the one-step target and TD error are

$$Y_t^Q = R_{t+1} + \gamma(1 - D_{t+1}) max_{a'} Q(S_{t+1}, a'), \tag{20}$$

$$\delta_t^Q = Y_t^Q - Q(S_t, A_t), \tag{21}$$

where $D_{t+1} = 1$ indicates termination and prevents value from leaking beyond the episode. The tabular update is

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha \delta_t^Q. \tag{22}$$

Thus $\alpha \in (0, 1]$ controls how far the old estimate moves toward the new sample target. The update is a stochastic approximation to equation (19) [8].

Q-learning is *off-policy*: the maximum in equation (21) evaluates a greedy continuation even when the behavior policy chose a non-greedy action. In our epsilon-greedy behavior policy,

$$\pi_\epsilon(a|s) = \begin{cases} 1 - \epsilon + \epsilon/|\mathcal{A}|, & a \in \arg\max_b Q(s, b) \quad \text{(with tie handling)}, \\ \epsilon/|\mathcal{A}|, & \text{otherwise.} \end{cases} \tag{23}$$

Large early ϵ discovers interventions whose long-term value was initially unknown; decay later exploits accumulated knowledge. Too little exploration leaves goals unvisited, whereas too much late exploration obscures competence and lengthens paths.

For a finite tabular MDP, bounded rewards, sufficient visitation, and learning rates satisfying standard stochastic-approximation conditions, Q-learning converges to Q^* with probability one [8]. Fixed learning rates, finite data, partial observability, and nonlinear recurrent approximation weaken these guarantees. Our results are therefore empirical and must be checked across held-out episodes and seeds.

3.4 Goal conditioning and sparse success signals

Ordinary RL learns one reward problem. Here a single agent faces a family of goals $g \in \mathcal{G}$. Goal conditioning augments the information state and value function:

$$Q^\pi(s, g, a) = \mathbb{E}_\pi[G_t \mid S_t = s, G = g, A_t = a]. \tag{24}$$

This permits experience about common dynamics to be shared while retaining different policies for different aims. It also makes geometry meaningful: the goals are compared through behavior learned in the same environment and model, rather than through unrelated specialist agents.

Sparse terminal rewards are conceptually clean but hard to discover by random exploration. The sequence experiments therefore add checkpoint rewards and a small per-step penalty. Reward shaping changes the finite-data learning problem, so success and unshaped intervention counts are still evaluated directly. In the spatial experiment all goals have the same terminal form and the central geometric quantity is not discounted return; it is empirical first-passage cost after training.

3.5 Partial observability, beliefs, and histories

In a partially observable MDP (POMDP), the environment has a latent Markov state X_t, but the agent observes only O_t generated from it [9]. The exact Bayesian information state is the belief

$$b_t(x) = \mathbb{P}(X_t = x \mid h_t), \tag{25}$$

which is a probability distribution over hidden states. Given a known model, the belief update combines prediction under the chosen action with Bayes' rule. In our setting the quantum state and instrument are unknown, so the agent cannot perform that privileged update. It must act from history, $\pi(a_t|h_t, g)$, or learn a compressed substitute.

The finite-history baseline uses

$$s_t^{(L)} = (g, j_t, (a_{t-L}, o_{t-L}), \dots, (a_{t-1}, o_{t-1})), \tag{26}$$

padding short histories at episode start. This representation is transparent and works well when the relevant memory is genuinely short. Its number of possible suffixes grows roughly as $(|\mathcal{A}||\mathcal{O}|)^L$, however, and two histories share statistical strength only if their suffixes are identical. A long literal suffix can also be *too discriminating*: it may treat two routes to the same operational place as different states. That issue becomes decisive in section 7.15.

3.6 Predictive state as an operational equivalence class

Predictive-state representations replace beliefs about a privileged latent variable by predictions of observable future tests [10]. A test $q = (a_1, o_1, \dots, a_k, o_k)$ asks for the probability that outcomes $o_{1:k}$ occur if interventions $a_{1:k}$ are imposed. Define

$$h \sim_{\text{pred}} h' \quad \iff \quad \mathbb{P}(q \mid h) = \mathbb{P}(q \mid h') \text{ for every feasible future test } q. \tag{27}$$

Histories in the same equivalence class have identical future affordances and need not be distinguished for prediction or control. This is precisely the agent-centered notion of state needed here: it is defined by possible future experience, not by access to ρ.

Exact predictive equivalence is demanding because there are arbitrarily long tests. We approximate it with a recurrent vector $z_t = F_\theta(h_t)$ trained jointly for reward and one-step outcome prediction. One-step accuracy alone does not prove sufficiency for long-horizon control; it is an inductive bias and diagnostic. Stronger future work should test multi-step predictions and whether histories with similar z remain interchangeable under novel intervention sequences.

3.7 GRU memory and function approximation

A recurrent network recursively maintains fixed-size memory

$$z_{t+1} = F_\theta(z_t, a_t, o_t), \qquad z_0 = \mathbf{0}. \tag{28}$$

With x_t the learned embedding of the action/outcome pair, a gated recurrent unit (GRU) [11] can be written

$$r_t = \sigma(W_r x_t + U_r z_t + b_r), \qquad u_t = \sigma(W_u x_t + U_u z_t + b_u), \tag{29}$$

$$\widetilde{z}_{t+1} = \tanh(W_h x_t + U_h(r_t \odot z_t) + b_h), \qquad z_{t+1} = (1 - u_t) \odot \widetilde{z}_{t+1} + u_t \odot z_t. \tag{30}$$

Here $\sigma(v) = 1/(1 + e^{-v})$ is applied coordinatewise and $\odot$ denotes coordinatewise multiplication. The reset gate r_t controls which previous features enter the candidate memory; the update gate u_t

interpolates between old and candidate memory. Unlike a fixed suffix, the same vector can learn to retain an old decisive outcome while forgetting many irrelevant recent events.

The value table is replaced by a neural function $Q_\theta(z, g, j, a)$. Given target Y_t^Q, the squared TD loss is

$$\mathcal{L}_Q(\theta) = \frac{1}{B} \sum_{t \in \mathcal{B}} [Y_t^Q - Q_\theta(z_t, g, j_t, a_t)]^2. \tag{31}$$

The target depends on a second parameter vector θ^-, held fixed while differentiating the loss, because otherwise the network would chase a target that moves in the same gradient step. Following deep Q-learning practice [12], it is updated softly:

$$\theta^- \leftarrow (1 - \tau)\theta^- + \tau\theta. \tag{32}$$

Complete episodes are unrolled and replayed so losses at late time steps can alter earlier memory updates through backpropagation through time. Gradient clipping limits rare unstable updates. None of these devices restores the tabular convergence theorem; they are practical stabilizers.

3.8 How an RL claim should be evaluated

An RL result combines at least three random sources: stochastic environment outcomes, exploratory action choices, and parameter initialization or data order. Training also changes its own data distribution. Consequently, a single learning curve is not a stable estimate of an algorithm's expected performance. Reproducibility studies document ranking reversals and unreliable uncertainty estimates from small seed counts [17, 18].

We separate the following questions.

Did the policy achieve the task?
: Held-out success probability estimates $\mathbb{P}_\pi(\tau_g \leq T)$. It is the primary competence measure.

How quickly did successful trials finish?
: Conditional mean duration estimates $\mathbb{E}[\tau_g \mid \tau_g \leq T]$. It must never be read alone, because a policy that succeeds only on easy random outcomes can look artificially fast.

How does reliability accumulate with time?
: The curve $T \mapsto \mathbb{P}(\tau_g \leq T)$ reveals whether two policies with the same eventual success differ at useful deadlines.

Did a representation acquire the intended meaning?
: Embeddings are compared with held-out policy and trajectory distances rather than judged by appearance. Learned costs are compared with empirical hitting times rather than accepted as distances by construction.

Does the conclusion survive nuisance variation?
: We report environments, goals, and random seeds separately before aggregates. Sample standard deviations describe the observed runs but, with two or three seeds, do not provide precise population uncertainty.

Finite horizons create right censoring: failure tells us only that $\tau_g > T$, not its actual value. One response is a restricted mean $\mathbb{E}[\min(\tau_g, T)]$, which keeps failures in the quantity but depends on the chosen horizon. Another is to report the entire success curve. We use both ideas where appropriate and explicitly label conditional quantities. These evaluation distinctions will later prevent geometric quality from being confused with mere navigation success.

4 Sequence goals and three geometries

The word *geometry* is used in three related but nonidentical senses. A strategic geometry compares two goals with each other. A trajectory geometry compares the long-run behaviors used to pursue them. A reachability geometry compares the current operational state with each goal. The first two are symmetric dissimilarities; the third is naturally directed. Keeping these objects separate prevents a common category error: two goals can require similar actions without either being easy from the present state.

4.1 Goals as finite-state recognition problems

A sequence goal is an ordered list of action/outcome checkpoints

$$g = ((a_1^g, o_1^g), \ldots, (a_{m_g}^g, o_{m_g}^g)). \tag{33}$$

Other events may occur between checkpoints. Progress $J_t \in \{0, \ldots, m_g\}$ is the number of checkpoints already matched. It updates deterministically:

$$J_{t+1} = J_t + mathbb1\{(A_t, O_t) = (a_{J_t+1}^g, o_{J_t+1}^g)\}, \tag{34}$$

with $J = m_g$ absorbing. Thus each goal defines a small deterministic automaton driven by observable experiences. Including J_t in the agent's input does not leak quantum state; it is a lossless computation from history and the stated task.

Rewards are 10 for completion, 1 for an intermediate checkpoint, and -0.01 otherwise. The completion bonus emphasizes success, checkpoint reward helps credit assignment, and step penalty weakly prefers shorter solutions. A goal-conditioned value $Q(z, g, j, a)$ lets the same remembered history support different actions for different desired futures. The spatial study later uses a simpler family of one-checkpoint goals, all expressed through the same probe action.

4.2 What properties make a distance?

A mathematical metric d obeys nonnegativity, identity of indiscernibles, symmetry, and the triangle inequality:

$$d(x,y) \geq 0, \quad d(x,y) = 0 \Leftrightarrow x = y, \quad d(x,y) = d(y,x), \quad d(x,z) \leq d(x,y) + d(y,z). \tag{35}$$

Operational goal costs need not satisfy every axiom. Distinct goals can be behaviorally interchangeable; quantum backaction can violate symmetry; finite sampling and imperfect policies can violate the triangle inequality. We use *geometry* broadly for structured relations among goals, while reserving *Euclidean-compatible metric* for a symmetric matrix that is well represented by ordinary Euclidean distances. Section 7 subjects that stronger claim to explicit tests.

4.3 Symmetric strategic similarity

Each known goal has embedding $e_g \in \mathbb{R}^k$. Distance $D_E(g,h) = \|e_g - e_h\|_2$ is intended to mean similarity of strategy, not labels. At encountered state z, define

$$\pi_g(a|z,j) = \mathrm{softmax}_a(Q(z,g,j,a)/T). \tag{36}$$

For policies p, q, Jensen–Shannon divergence is

$$D_{\mathrm{KL}}(p\|q) = \sum_a p(a) \log \frac{p(a)}{q(a)}, \qquad D_{\mathrm{JS}}(p,q) = \tfrac{1}{2} D_{\mathrm{KL}}(p\|m) + \tfrac{1}{2} D_{\mathrm{KL}}(q\|m), \quad m = \tfrac{1}{2}(p+q). \tag{37}$$

Mixing through m keeps the Jensen–Shannon divergence finite when one policy assigns zero probability to an action favored by the other. It is symmetric, bounded, and its square root is a metric [14]. If two goals induce nearly identical soft action preferences at many encountered states, they are strategically close even when their textual labels differ. The regularizer is

$$\mathcal{L}_{\text{geom}} = \frac{1}{|\mathcal{P}|} \sum_{g<h} [\|e_g - e_h\|_2 - c\sqrt{D_{\text{JS}}(\pi_g, \pi_h)}]^2. \tag{38}$$

where c sets units and $\mathcal{P}$ is a sampled set of goal pairs. The policy target is detached: gradients change embeddings to reflect current policy dissimilarity but do not distort the policy merely to make the geometry easy. Geometry is nevertheless partly imposed by this loss. Agreement on held-out states tests whether that inductive bias captures persistent strategy; it is not evidence of spontaneous emergence without bias.

4.4 Trajectory similarity

Local policy similarity may miss long-horizon differences. From held-out rollouts we build

$$\phi(g) = [\widehat{f}_g(a), \widehat{f}_g(a, o), \widehat{\mathbb{E}}(\tau_g/T), \widehat{\mathbb{P}}(\tau_g \leq T)] \tag{39}$$

and $D_{\text{traj}}(g, h) = \|\phi(g) - \phi(h)\|_2$. This resembles successor features, which describe policies by expected future features and support transfer [15], although our signature is diagnostic rather than a learned successor model.

The distinction is important. D_{policy} compares action distributions at a common collection of local states. D_{traj} compares the visitation distributions those policies actually induce over complete rollouts. Two goals may choose the same first action and then diverge; conversely, different local choices may lead to similar aggregate frequencies. Correlation with both quantities is stronger evidence than correlation with the training target alone.

4.5 Directed reachability and displacement

Let τ_g be first completion time. The ideal intervention-count distance is

$$\tau_g = \inf\{k \geq 0 : J_{t+k} = m_g\}, \qquad d^*(z, g, j) = \min_\pi \mathbb{E}_\pi[\tau_g | z, j]. \tag{40}$$

This is a stochastic shortest-path problem [13] with Bellman form

$$d^*(z, g, j) = \begin{cases} 0, & j = m_g, \\ 1 + \min_a \sum_o P(o|z, a) d^*(F(z, a, o), g, j'(a, o)), & j < m_g. \end{cases} \tag{41}$$

Every nonterminal intervention contributes one unit; the expectation averages over measurement outcomes; and minimization chooses the best first action. Unlike discounted return, this quantity is expressed directly in interventions. It can be infinite when a goal is unreachable, which is why finite-horizon empirical estimates require careful failure handling.

The network learns positive action cost $C(z, g, j, a)$ and sets $d(z, g, j) = \min_a C(z, g, j, a)$. Its target is 1 on completion and otherwise

$$y_t^C = 1 + \min_{a'} C_{\theta^-}(z_{t+1}, g, j_{t+1}, a'). \tag{42}$$

This target bootstraps exactly as Q-learning does, but minimizes positive cost rather than maximizing reward. It is directed: measurement can destroy conditions needed to return. It is also policy-sensitive in finite data, because samples come from reward-oriented exploratory behavior rather than a separate cost-optimal policy.

For N goals, the agent's goal-relative position is

$$\boldsymbol{D}(z) = (d(z, g_1), \ldots, d(z, g_N)). \tag{43}$$

An observed intervention produces displacement $\Delta \boldsymbol{D}_t = \boldsymbol{D}(z_{t+1}) - \boldsymbol{D}(z_t)$. Negative coordinates mean closer, positive farther. With the learned predictor, a contemplated action has

$$\mathbb{E}[d'_g|z, a] = \sum_o \widehat{P}(o|z, a) d(F(z, a, o), g), \tag{44}$$

which provides a route to model-based planning.

The vector $\boldsymbol{D}(z)$ behaves like an agent-centered coordinate chart: it says where the present lies relative to a fixed landmark set of goals. These are not Cartesian coordinates. They can be redundant, curved, asymmetric, and high-dimensional. If sufficiently many landmark distances obey a common low- dimensional Euclidean model, one may subsequently recover Cartesian-like coordinates by multidimensional scaling. That additional step is the central subject of section 7.

A scalar mean hides risk, so we also estimate

$$R_g(z, T) = \mathbb{P}(\tau_g \leq T|z, \pi_g). \tag{45}$$

This parallels distributional RL's emphasis on distributions rather than only expected return [16].

For example, a deterministic three-step route and a one-step gamble that succeeds with probability one half and otherwise takes a long detour can have similar means but very different deadline reliability. A full first-passage distribution distinguishes them. The present work records empirical success curves; learning calibrated distributions directly is left to future work.

5 Architecture and experimental protocol

All agents share one interaction protocol and receive only goal ID, progress, action, outcome, reward, and termination. They never receive a density matrix, state fidelity, target-state overlap, Born probability, or hidden coordinate. This common interface makes architectural comparisons meaningful: differences come from memory and auxiliary learning objectives rather than extra physical information.

5.1 Three learner families

The **finite-history tabular agent** uses equation (26) with $L = 4$. It stores one scalar per state–action pair and applies equation (22). It is statistically efficient when the relevant suffixes are few and frequently revisited, interpretable because every entry can be inspected, and incapable of generalization between distinct keys.

The **reward-only GRU agent** applies equation (30) to the complete interaction sequence. At each decision it concatenates recurrent memory z_t, a learned goal embedding e_g, and one-hot goal progress, then feeds that vector to a multilayer perceptron producing one Q-value per legal action. The embedding is simply a trainable lookup vector in this baseline; no geometric loss constrains it.

The **predictive geometry GRU** shares the same recurrent trunk but has four coupled outputs: reward value, one-step outcome prediction, positive cost-to-go, and goal embedding. Parameter sharing asks the memory to retain features useful across control, prediction, and reachability. It is a multi-task inductive bias, not a guarantee that a unique latent state will be identified.

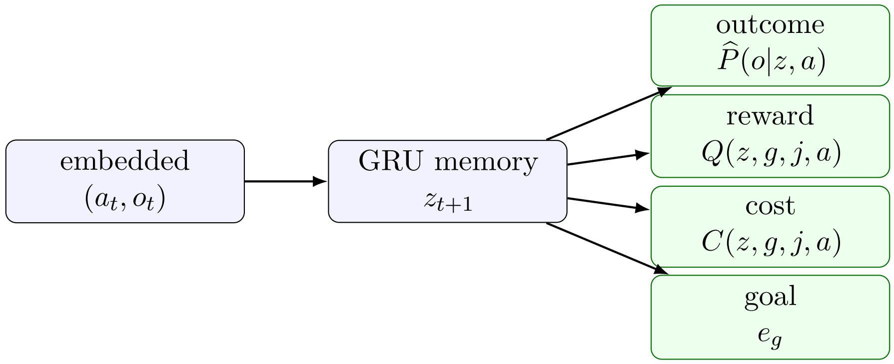


Figure 2: Shared recurrent representation and four learning objectives. Outcome prediction is goal-independent; reward and cost are goal-conditioned.

5.2 Auxiliary objectives and their roles

Before the current outcome is incorporated, the outcome head predicts $\widehat{P}_\theta(o_t|z_t, a_t)$. Its categorical negative log likelihood is

$$\mathcal{L}_{\text{pred}} = -\log \widehat{P}_\theta(o_t|z_t, a_t), \tag{46}$$

with impossible outcomes masked from the softmax. Predicting after ingesting o_t would permit trivial copying; the temporal ordering is therefore part of the scientific validity of the auxiliary task.

For the selected action, reward and cost losses are squared temporal- difference residuals,

$$\mathcal{L}_Q = (Y_t^Q - Q_\theta(z_t, g, j_t, a_t))^2, \tag{47}$$

$$\mathcal{L}_C = (Y_t^C - C_\theta(z_t, g, j_t, a_t))^2. \tag{48}$$

The cost output is transformed to remain positive. The geometry loss is equation (38). The full per-batch objective is

$$\mathcal{L} = \mathcal{L}_Q + \lambda_P \mathcal{L}_{\text{pred}} + \lambda_C \mathcal{L}_C + \lambda_G \mathcal{L}_{\text{geom}}. \tag{49}$$

The coefficients are necessary because the losses have different numerical scales. A large prediction weight can improve next-outcome modeling while diverting capacity from reward; a large geometry weight can make embeddings faithful to a transient, inaccurate policy. We therefore treat the full model as a testable architectural proposal rather than presuming that more auxiliary objectives must improve control.

Goals are sampled uniformly. Each episode resets hidden state, memory, and progress. Trajectories are replayed once for backpropagation through time; reward and cost use equation (32); gradients are clipped to norm 5.

Uniform goal sampling makes the optimization objective a macro-average over the declared repertoire rather than a frequency-weighted average favoring easy or common goals. At each environment step, action masks prevent selecting an outcome logit or action that does not exist in a variable-alphabet instrument. Random-number generators for environment, exploration, and neural initialization are seeded by run; evaluation uses separate deterministic seed streams so that training and test trajectories do not overlap.

Evaluation disables exploration. Separate geometry rollouts evaluate every goal-conditioned policy at each state, build trajectory signatures, compare blank-history cost to hitting time, estimate equation (45), and group displacements by action/outcome. PCA is used only to visualize embeddings; all distance comparisons use original dimensions.

Table 2: Production configuration.

Training episodes/run	400	Evaluation episodes/goal	100
Geometry episodes/goal	40	Horizon	15
Discount γ	0.95	Tabular α	0.10
History length	4	Neural learning rate	10^{-3}
ϵ start/end/decay	1/.05/.99	Target τ	.02
GRU/goal dimensions	32/6	Interaction embedding	8
$\lambda_P, \lambda_C, \lambda_G$	.5, .5, .05	Geometry scale	2
Weak strength	.8	Seeds	0,1

PCA deserves a caution. It selects orthogonal directions of greatest sample variance in e_g, not directions that best preserve pairwise distance or control meaning. A two-dimensional PCA panel is therefore an illustration. Reported embedding correlations always use the original six-dimensional distances. The later spatial study instead uses MDS because its scientific question is explicitly whether pairwise costs admit low-dimensional Euclidean coordinates.

The study contains 14,400 training episodes (107,804 interventions), 26,400 evaluation episodes (179,342 interventions), and 3,520 geometry episodes. An exact manifest, tidy CSVs, model weights, and plots are saved. Eighteen tests verify quantum validity, outcome masks, reproducibility, goals, runner semantics, every agent, and geometry utilities.

6 Results

6.1 Control performance

Control competence must be established before representation quality is interpreted. An embedding learned by an agent that seldom reaches its goals mostly describes erroneous action values. We therefore begin with held-out greedy success and only then ask whether internal quantities have geometric meaning. Success is averaged equally over goals within a scenario and then equally over scenarios for the macro-average; this avoids allowing an environment with more goals to dominate the headline number.

No backend dominates. Recurrent memory helps on the qutrit and original $|1\rangle$ scenarios; tabular learning leads on Pauli, unsharp, and Pauli+SIC; the full model is strongest from $|+\rangle$. Macro conditional steps are 4.62 (tabular), 4.55 (GRU), and 4.41 (full), but success remains primary.

The macro differences are small: the full model exceeds tabular success by 1.4 percentage points and the plain GRU by 2.8 points. These values do not justify a universal superiority claim. They show that adding predictive, cost, and geometry objectives is compatible with competitive control at this scale. The scenario-level reversals are scientifically more informative. Short finite histories are a strong bias in small measurement worlds, where the same exact suffix recurs often. Recurrent compression becomes more useful as the qutrit alphabet and possible histories grow, but its larger parameter space also increases optimization variance.

Conditional intervention counts answer a secondary question: among trials that finish, how direct are the successful paths? The full agent's value of 4.41 is numerically lowest, but it cannot compensate for a lower success rate on a particular task. For illustration, a policy that succeeds in one step on 10% of trials and fails otherwise has conditional duration one, yet is worse for most purposes than a policy that reliably succeeds in five steps. This is why figure 3 juxtaposes rather than combines the two metrics.

Table 3: Success rate, mean ± sample standard deviation over two seeds. Bold is the largest scenario mean.

World / state	Finite history	GRU	Predictive geometry GRU
Z/X/weak-Z / $\|1\rangle$	$.810 \pm .012$	$\mathbf{.874 \pm .038}$	$.861 \pm .035$
Z/X/weak-Z / $\|+\rangle$	$.878 \pm .009$	$.809 \pm .063$	$\mathbf{.924 \pm .005}$
Pauli / $\|+i\rangle$	$\mathbf{.863 \pm .018}$	$.806 \pm .057$	$.825 \pm .169$
unsharp / $I/2$	$\mathbf{.849 \pm .008}$	$.788 \pm .080$	$.820 \pm .071$
Pauli+SIC / $I/2$	$\mathbf{.713 \pm .006}$	$.649 \pm .027$	$.697 \pm .172$
qutrit MUB / $\|2\rangle$	$.634 \pm .034$	$\mathbf{.734 \pm .082}$	$.703 \pm .015$
Macro-average	.791	.777	**.805**

Neural instability is substantial. Mean absolute seed gaps are 2.1 percentage points for tabular, 8.2 for GRU, and 11.0 for the full model; the largest full gap is 24.3. Both recurrent agents have zero success on the three-checkpoint qutrit goal, while tabular reaches 15%. The plain GRU also has zero success on the two-step $Z0\_SIC0$ goal.

These failures are consistent with a sparse-credit problem. A long sequence goal is completed only when several ordered events occur within a short horizon; early in training, its terminal bonus is rarely observed. The tabular agent can preserve a lucky exact suffix once encountered, whereas the recurrent learner must alter a distributed representation without corrupting other goals. More episodes, curriculum learning, eligibility traces, or demonstrations could change the ranking. The present comparison evaluates the specified finite-budget learners, not their asymptotic capacities.

6.2 Goal-geometry validation

The embedding loss explicitly encourages policy similarity, so training-loss fit would be circular evidence. We instead compute pairwise distances from held-out greedy behavior. Spearman rank correlation is chosen because the embedding scale is arbitrary and the most defensible first claim is ordinal: goals represented as nearer should generally require more similar strategies. If r_i and s_i are the ranks of two lists of pairwise distances, Spearman correlation is simply their Pearson correlation,

$$\rho_S = \frac{\sum_i (r_i - \bar{r})(s_i - \bar{s})}{\sqrt{\sum_i (r_i - \bar{r})^2 \sum_i (s_i - \bar{s})^2}}. \tag{50}$$

It is invariant under monotone rescaling and therefore does not claim metric calibration.

Across 12 full-agent runs, embedding distance has mean Spearman correlation 0.537 with held-out policy strategy distance (median 0.533, standard deviation 0.207, range 0.121–0.825). This is meaningful but imperfect generalization of the regularized relation. Correlation with more independent trajectory distance is weaker: mean 0.311, median 0.274, standard deviation 0.303, range −0.091–0.786. Geometry is not yet a unique robust object. The first two PCA axes explain 77.3% of embedding variance on average.

The gap between policy and trajectory correlations has a useful interpretation. The regularizer observes local soft policies at sampled states, so $\rho = 0.537$ indicates moderate out-of-sample persistence of its intended relation. Trajectory signatures additionally incorporate the states the policy visits, stochastic outcomes, duration, and success. Their weaker $\rho = 0.311$ shows that similar local preferences do not yet imply reliably similar global behavior. Negative correlations in some seeds establish that the embedding is not a seed-invariant intrinsic geometry.

Explained PCA variance is reported only to quantify how much of the learned six-dimensional lookup table can be displayed on a page. A value of 77.3% does not establish a two-dimensional

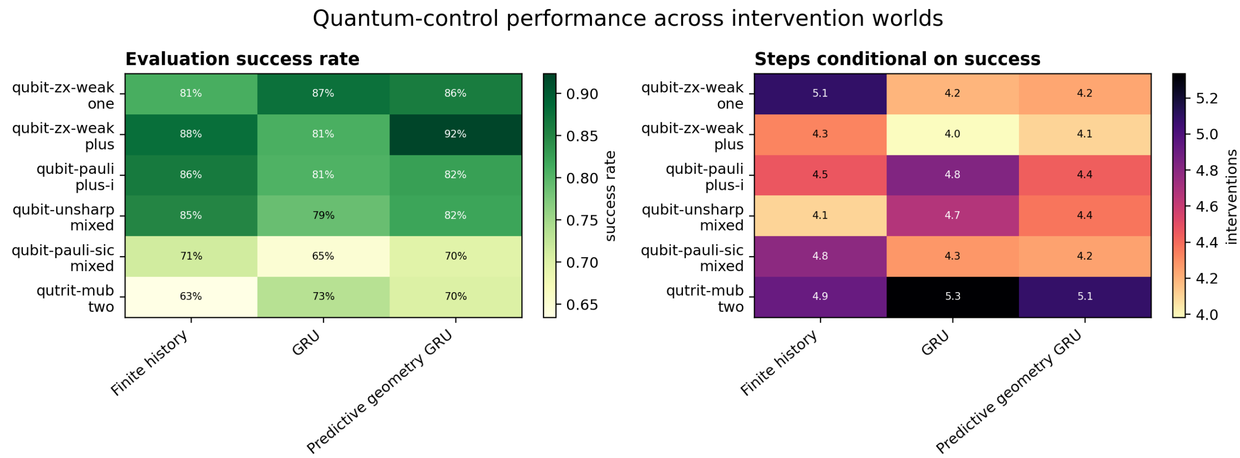


Figure 3: Held-out success and conditional intervention count, averaged over two seeds. Conditional steps exclude failures and must be read beside success.

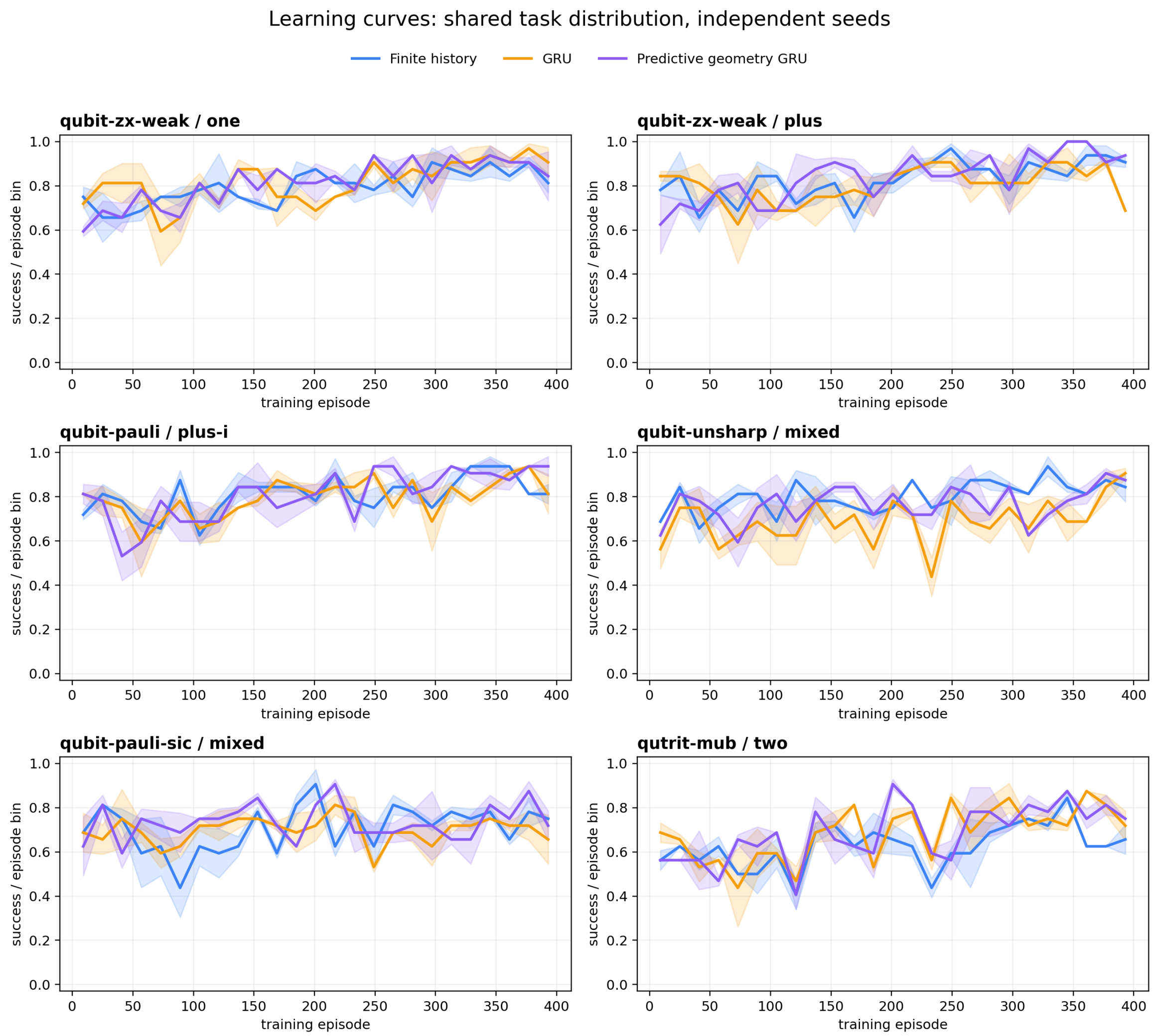


Figure 4: Binned exploratory training success. Shading is standard error over only two seeds and is descriptive, not a reliable interval.

manifold: PCA preserves variance rather than all pairwise distances, the goal set is small, and the embeddings were not trained to produce spatial coordinates. The inverse-designed study adopts much stricter dimensional diagnostics.

6.3 Reachability and intervention displacement

Reachability evaluation asks a more demanding question than rank-order similarity: does a learned scalar equal an empirical number of interventions? We compare the blank-history prediction for each goal with mean hitting time on successful geometry rollouts. Pearson correlation measures linear ordering and mean absolute error (MAE)

$$\mathrm{MAE} = \frac{1}{M}\sum_{i=1}^{M} |\widehat{d}_i - \widehat{\mathbb{E}}(\tau_i \mid \tau_i \le T)| \tag{51}$$

measures calibration in intervention units. Correlation can be high despite a wrong scale; low MAE can arise from predicting the mean for a narrow range. Both are therefore needed.

Across 79 goals with at least one geometry-rollout success, predicted cost correlates $r = 0.471$ with conditional empirical steps and has mean absolute error 1.75. Predictions are compressed to 2.46–5.89 while observed means span 1.0–10.71. The head contains ordinal information but is not a calibrated clock. Off-policy minimum backups, function approximation, policy mismatch, finite-horizon censoring, and conditioning on success all contribute.

The compression of predictions is visible evidence of regression toward frequently observed targets: very easy and very hard goals are pulled toward a middle range. Off-policy minimum backups add downward selection bias because a minimum over noisy action estimates tends to choose an underestimated value. At the other side, failed trials are absent from the conditional empirical mean. The comparison is therefore deliberately diagnostic rather than a claim that either axis is an uncensored ground truth. The success curves in figure 7 preserve the missing reliability information.

In figure 8, Z:0, X:0, and Y:0 change matching-goal distance by -1.58, -1.80, and -1.91. The four SIC outcomes change their matching distances by -2.43 to -2.95, while most nonmatching coordinates increase. Completion explicitly sets a matching goal to zero, so diagonals should not be overinterpreted; structured off-diagonal change is the more interesting signature of backaction.

An off-diagonal cell answers a counterfactual-geometric question: after experiencing outcome o of action a, did an unrelated goal become easier or harder according to the same predictive state? A coherent row pattern means the representation tracks a joint reorganization of affordances rather than only recognizing terminal reward. Because these values come from the learned cost head, they remain hypotheses about operational displacement. Their next validation should compare predicted and empirical post-intervention success curves for every goal.

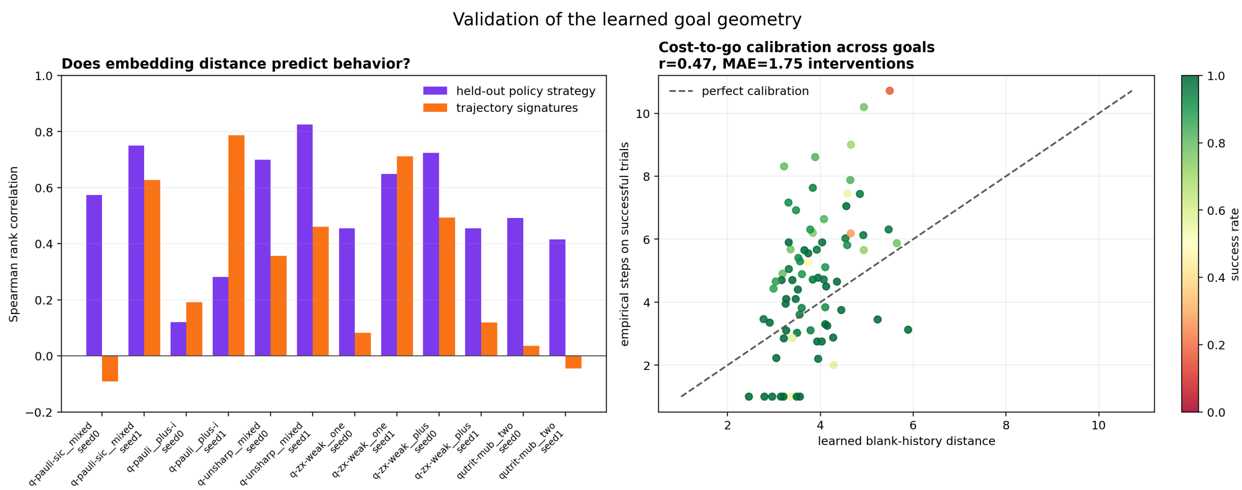


Figure 5: Left: correlation of embedding distance with held-out policy and trajectory distances. Right: cost-to-go calibration across successful goals.

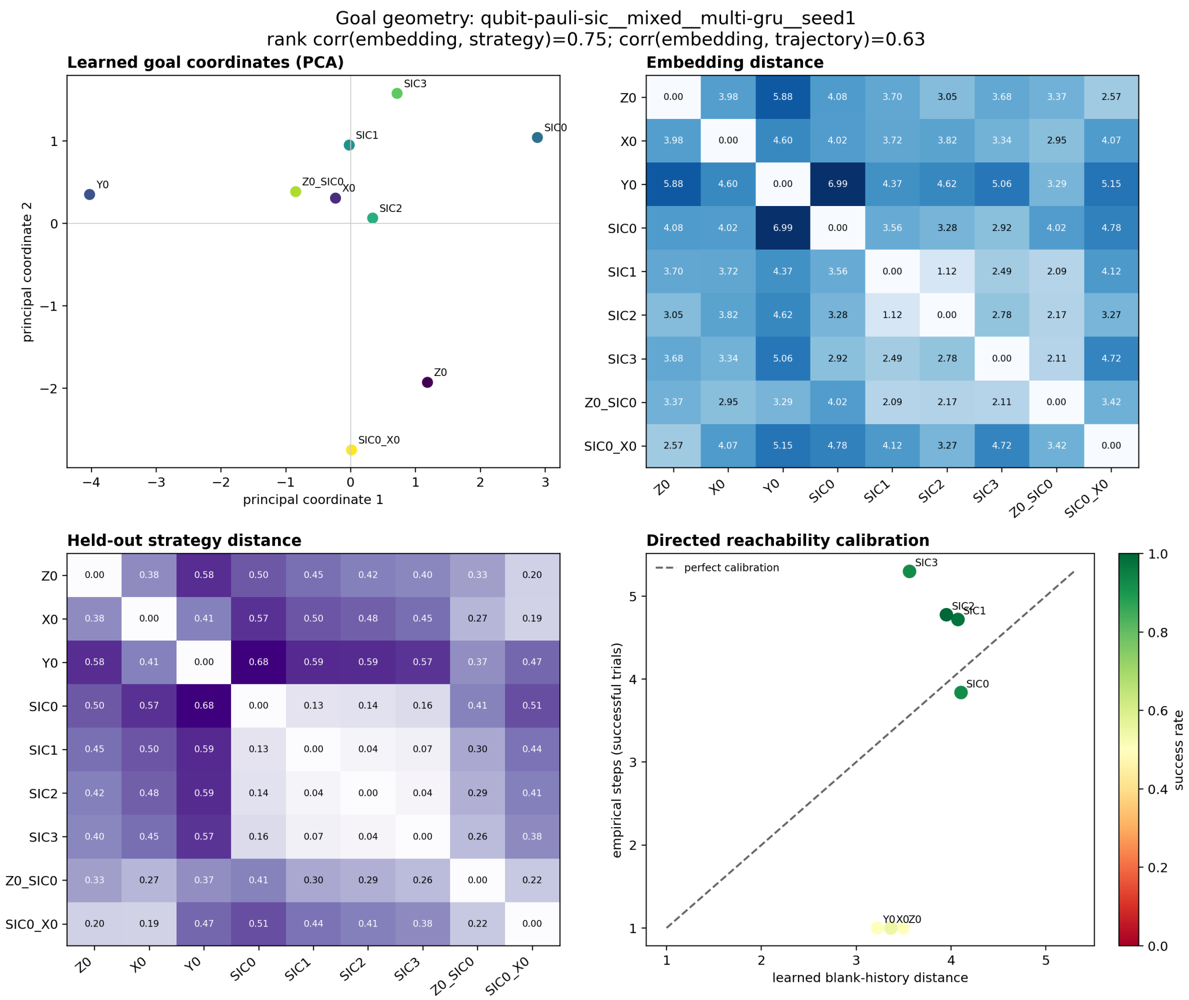


Figure 6: Representative Pauli+SIC geometry (seed 1): PCA coordinates, full embedding distances, held-out policy distances, and reachability calibration. The embedding has strategy correlation 0.75 and trajectory correlation 0.63 in this run.

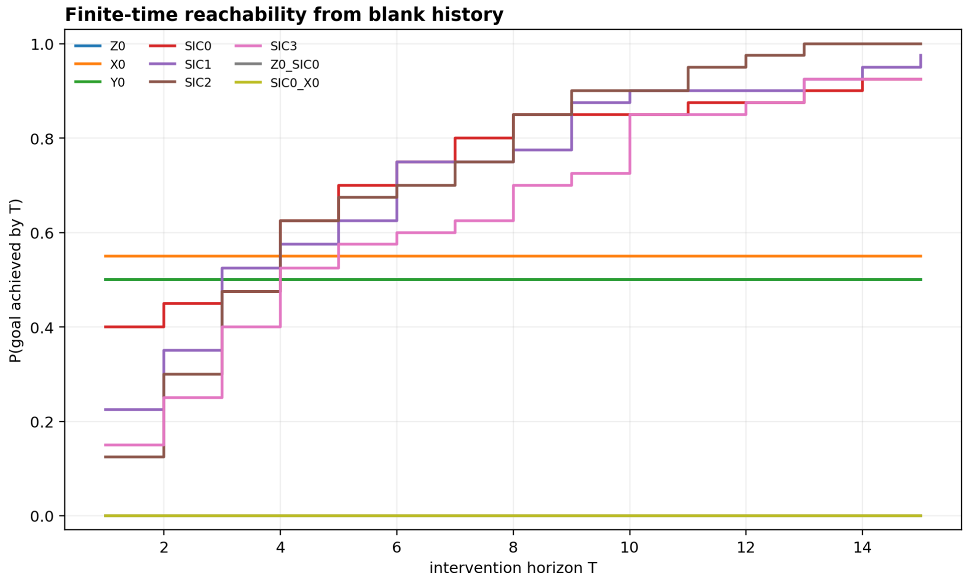
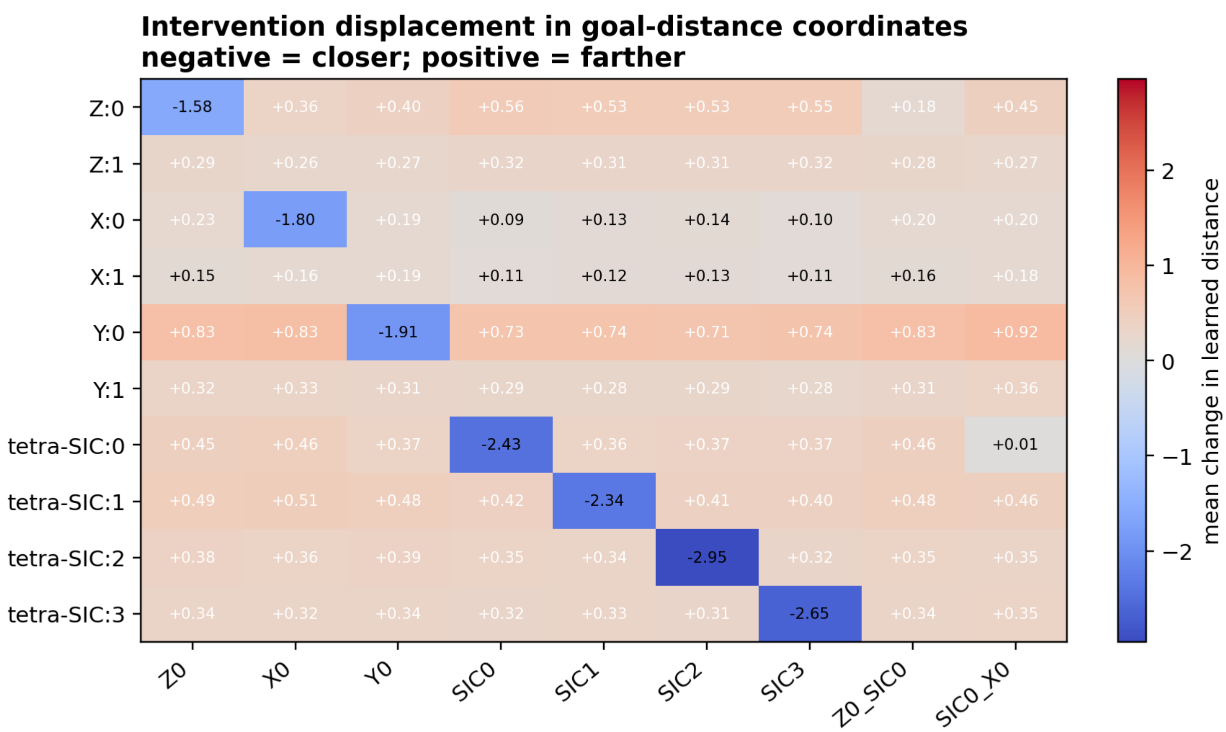


Figure 7: Finite-time success curves retain reliability information hidden by a conditional mean hitting time.



	Z0	X0	Y0	SIC0	SIC1	SIC2	SIC3	Z0_SIC0	SIC0_X0
Z:0	-1.58	+0.36	+0.40	+0.56	+0.53	+0.53	+0.55	+0.18	+0.45
Z:1	+0.29	+0.26	+0.27	+0.32	+0.31	+0.31	+0.32	+0.28	+0.27
X:0	+0.23	-1.80	+0.19	+0.09	+0.13	+0.14	+0.10	+0.20	+0.20
X:1	+0.15	+0.16	+0.19	+0.11	+0.12	+0.13	+0.11	+0.16	+0.18
Y:0	+0.83	+0.83	-1.91	+0.73	+0.74	+0.71	+0.74	+0.83	+0.92
Y:1	+0.32	+0.33	+0.31	+0.29	+0.28	+0.29	+0.28	+0.31	+0.36
tetra-SIC:0	+0.45	+0.46	+0.37	-2.43	+0.36	+0.37	+0.37	+0.46	+0.01
tetra-SIC:1	+0.49	+0.51	+0.48	+0.42	-2.34	+0.41	+0.40	+0.48	+0.46
tetra-SIC:2	+0.38	+0.36	+0.39	+0.35	+0.34	-2.95	+0.32	+0.35	+0.35
tetra-SIC:3	+0.34	+0.32	+0.34	+0.32	+0.33	+0.31	-2.65	+0.34	+0.35

Figure 8: Mean displacement in learned goal-distance coordinates. Blue means closer and red farther. Matching outcomes create strong negative diagonals, while backaction alters multiple alternative affordances.

7 Inverse-designing emergent two-dimensional space

7.1 From a philosophical proposal to an empirical criterion

The preceding experiments ask whether a finite goal repertoire acquires useful structure. The stronger hypothesis motivating this project is that ordinary space itself may be a special form of an agent's goal geometry: goals that are easy to achieve are near, goals that require many interventions are far, and a policy is motion through the resulting hodological space. This statement is not yet an experiment. To make it falsifiable, we need operational definitions of place, distance, dimension, and trajectory that do not silently provide the answer to the learner.

Suppose there are n repeatably preparable operational situations $s_1, \ldots, s_n$ and n corresponding place goals $g_1, \ldots, g_n$. The ordered policy cost from source i to goal j is

$$C_{ij}^{\pi} = \mathbb{E}_{\pi_j}[\min(\tau_{g_j}, T) \mid s_i], \tag{52}$$

where T is the evaluation horizon. In a fully successful regime this reduces to expected hitting time. The reported spatial dissimilarity removes the one intervention that every goal uses for terminal verification:

$$D_{ij} = C_{ij}^{\pi} - c_{\text{probe}}, \qquad c_{\text{probe}} = 1. \tag{53}$$

Subtracting this common overhead is analogous to removing a fixed measurement latency from every journey. Without it, even D_{ii} would be one although no movement is needed. The subtraction affects absolute scale but not the learned policy.

A finite goal system supports an ordinary spatial interpretation only if several conditions hold together:

1. *competence*: policies reliably reach every goal from every source;
2. *localization*: histories contain enough operational information to select source-dependent actions, without being handed coordinates;
3. *metric coherence*: pairwise costs are approximately symmetric and obey a common low-distortion distance model;
4. *dimensional preference*: two dimensions fit much better than one, while three add little;
5. *trajectory coherence*: successive operational observations map to local paths in the coordinates derived from all-pairs cost;
6. *robustness*: these conclusions persist across learned policies, random seeds, and controls designed to remove specific ingredients.

No single item is sufficient. A perfect navigator can inhabit a non-Euclidean metric; a beautiful embedding can summarize failed policies; and a hidden grid can be invisible to an agent that cannot localize itself.

7.2 The inverse-design problem

Let θ parameterize the initial-state ensemble, quantum instruments, and observation coarse-graining, and let $\mathcal{G}$ denote the goal repertoire. Training algorithm $\mathcal{A}_{\text{RL}}$ maps interaction data to policies $\pi_{\theta,\mathcal{G}}$. The outer scientific problem is therefore bilevel:

$$\pi_{\theta,\mathcal{G}} = \mathcal{A}_{\text{RL}}(\mathcal{E}_\theta, \mathcal{G}), \tag{54}$$

$$(\theta^\star, \mathcal{G}^\star) = \arg\min_{\theta,\mathcal{G}} J_d\Big(D_{\pi_{\theta,\mathcal{G}}}\Big)\,. \tag{55}$$

The inner learner turns the proposed world into behavior; evaluation turns behavior into a cost matrix; the outer objective judges whether that matrix has the desired geometry. A useful general objective is

$$J_d = \lambda_s \,\mathrm{Stress}_d + \lambda_c P_{\text{calibration}} + \lambda_n P_{\text{nonmetric}} + \lambda_a P_{\text{asymmetry}} + \lambda_f P_{\text{failure}} + \lambda_k P_{\text{complexity}}. \tag{56}$$

The terms discourage different pathologies. Stress penalizes failure of a d-dimensional Euclidean fit. Calibration compares learned and reproducible costs. The nonmetric term detects dissimilarities that cannot be exact Euclidean distances even in higher dimension. Asymmetry detects irreversible or policy-biased travel. Failure prevents the optimizer from making difficult pairs disappear behind censoring. Complexity discourages a separate teleportation action for every goal or an unnecessarily large hidden model.

Optimizing general quantum channels through a stochastic RL loop is expensive and scientifically opaque as a first experiment. We instead choose a family whose exact stochastic shortest paths can be computed by a privileged outer loop, search one physically meaningful instrument parameter, and only then train coordinate-free agents. This separation makes the construction auditable: analytic intuition, exact design geometry, and learned empirical geometry can be compared rather than conflated.

7.3 Euclidean distance and the dimensional hypothesis

For candidate coordinates $x_i \in \mathbb{R}^d$, the Euclidean distance is $\delta_{ij}(X) = \|x_i - x_j\|_2$. Metric multidimensional scaling (MDS) asks for coordinates whose distances approximate the observed dissimilarities. We use Kruskal-style normalized raw stress [23]:

$$\mathrm{Stress}_d(D, X) = \sqrt{\frac{\sum_{i<j}[\delta_{ij}(X) - D_{ij}]^2}{\sum_{i<j} D_{ij}^2}}. \tag{57}$$

The numerator is total squared distance error; the denominator removes overall units. Stress zero means exact representation at the requested dimension. There is no universal threshold at which stress suddenly becomes acceptable: its interpretation depends on sample size, noise, and comparison conditions. We therefore emphasize relative gaps among dimensions and matched controls.

The dimensional hypothesis is not merely "2D stress is small." Any finite metric can often fit better when extra dimensions are added. The characteristic pattern sought here is

$$\mathrm{Stress}_1 \gg \mathrm{Stress}_2 \approx \mathrm{Stress}_3\,. \tag{58}$$

The first inequality rejects a line; the small second improvement says a third coordinate explains little additional structure. Larger studies should use held-out points or local dimension estimators to guard against overfitting. On nine exhaustively evaluated sites, the dimension curve and controls provide a transparent first test.

MDS is related in spirit to manifold learning, where global coordinates are reconstructed from local metric relations [27]. The important difference is the source of dissimilarity: our D_{ij} is not an externally measured physical separation. It is the intervention cost of a learned policy. The resulting axes consequently have no preferred orientation, handedness, origin, or unit until an external validation frame is applied.

7.4 Nine-dimensional quantum construction

7.4.1 Hilbert space, labels, and source ensemble

Let

$$\mathcal{H}_9 = \mathrm{span}\{|x, y\rangle : x, y \in \{0, 1, 2\}\}, \qquad \rho_0 = |1, 1\rangle\langle 1, 1|. \tag{59}$$

The nine basis vectors are mutually orthogonal localized states. The canonical episode starts at the central state ρ_0, but all-pairs training uniformly draws a source from the ensemble

$$\mathcal{S}_0 = \{|x, y\rangle\langle x, y| : x, y \in \{0, 1, 2\}\}. \tag{60}$$

This distinction is important. The central state answers the requested initial-state specification; random-source episodes teach translation of a local policy across the entire operational space. A geometry inferred only from the center would determine one row of D, not all pairwise relations.

The coordinate labels (x, y) exist only in environment construction, exact-cost calculation, and post hoc validation. The learner sees an arbitrary bijection to symbols A–I. Permuting these symbols before training changes no available statistical fact. Thus place identity is observable, but adjacency, axis, handedness, coordinate difference, and Euclidean distance are not.

7.4.2 Movement actions as quantum instruments

There are eight movement actions: north, south, east, west, and four diagonals. Let s abbreviate a basis site and let $f_a(s)$ be its neighboring destination when action a is legal. Their hidden source-resolved Kraus events are

$$K_{s,+}^{(a)} = \sqrt{p_a(s)}|f_a(s)\rangle\langle s|, \qquad K_{s,-}^{(a)} = \sqrt{1 - p_a(s)}|s\rangle\langle s|. \tag{61}$$

For a legal axial movement $p_a(s) = 1$; for a legal diagonal $p_a(s) = p_d$; and at an open boundary $p_a(s) = 0$ with $f_a(s)$ understood as s. The "+" branch moves, while the "−" branch remains in place.

Physical completeness follows explicitly. For each source s,

$$K_{s,+}^{(a)\dagger} K_{s,+}^{(a)} = p_a(s)|s\rangle\langle s|, \tag{62}$$

$$K_{s,-}^{(a)\dagger} K_{s,-}^{(a)} = [1 - p_a(s)]|s\rangle\langle s|. \tag{63}$$

Adding branches gives $|s\rangle\langle s|$; adding the nine orthogonal sources gives

$$\sum_s \sum_{b \in \{+,-\}} K_{s,b}^{(a)\dagger} K_{s,b}^{(a)} = \sum_s |s\rangle\langle s| = I_9. \tag{64}$$

Every action is therefore a valid trace-preserving quantum instrument for all $p_d \in [0, 1]$.

These channels also reveal the construction's present boundary. On arbitrary $\rho = \sum_{s,t} \rho_{st}|s\rangle\langle t|$, the source-resolved sum removes off-diagonal coherences because every term contains the same source projector on both sides. The movement is a measure-and-prepare, entanglement-breaking channel on the localized basis: a classical stochastic walk represented within quantum operations. This is a legitimate quantum environment and a clean existence proof, but it does not yet use coherent interference to generate the geometry.

7.4.3 Observed outcomes as a coarse-graining of hidden events

The environment need not report the hidden source index or branch. Several Kraus events can correspond to the same classical outcome, as in equation (3). In the main world, events are grouped by the place occupied after the action. The completely positive operation associated with observing destination y is

$$\mathcal{I}_y^{(a)}(\rho) = \sum_{s:f_a(s)=y} K_{s,+}^{(a)}\rho K_{s,+}^{(a)\dagger} + K_{y,-}^{(a)}\rho K_{y,-}^{(a)\dagger}. \tag{65}$$

where the first sum includes successful branches arriving at y, and the second term includes a failed attempt originating at y. Summing $\mathcal{I}_y^{(a)}$ over y recovers the trace-preserving action channel. Conditioned on observing y, the post-measurement state is $|y\rangle\langle y|$ whenever the input is a localized state. The observed symbol is therefore a place-cell-like localization event.

The blind control uses exactly the same hidden Kraus operators but groups events into only two observed outcomes: success and failure. It isolates the effect of observation: hidden connectivity and transition probabilities remain fixed while location becomes partially observed. Comparing the two asks whether an underlying spatial graph is enough when the agent cannot tell where it is.

7.4.4 A common probe and a homogeneous goal repertoire

A ninth action is a sharp place probe with projectors

$$P_s = |s\rangle\langle s|, \qquad \sum_s P_s^\dagger P_s = I_9. \tag{66}$$

The nine goals all have exactly the same syntax:

$$g_s = \text{“choose the common probe and observe symbol } s\text{”}. \tag{67}$$

There is no action called “go to A,” no coordinate-conditioned reward, and no separate detector dedicated to each target. A successful policy must use movement actions until its operational observation is appropriate and then use the common probe. Goal names distinguish desired outcomes but do not encode which actions approach them.

At the beginning of a random-source episode, the environment emits one ordinary probe result before the task clock starts. This supplies current place identity without charging a movement or verification cost. It reveals no neighbor table. From then onward, the agent must learn the transition affordances of each movement label through interaction. This arrangement cleanly separates *localization* (which place symbol am I at?) from *geometrization* (how are place symbols related by action cost?).

7.4.5 The coordinate-free Markov state

For the place-observed worlds, the observable decision state is

$$\widetilde{S}_t = (g, J_t, \ell_t), \tag{68}$$

where ℓ_t is the latest arbitrary place symbol. It deliberately discards the action and older history. This is not a hand-coded coordinate: it contains no pair (x, y) and no distance. It is a quotient of histories by current observable affordance. If two histories end with the same ℓ_t, every future action has the same outcome distribution, so retaining their different arrival routes would violate rather than improve Markov abstraction.

Tabular Q-learning over equation (68) must still discover which movement labels connect which symbols and how their stochastic costs compare. The blind world cannot use this state because the latest success/failure outcome does not identify place; it retains the longer finite-history learner. This matched difference becomes an observability ablation rather than a claim that memory is never necessary.

7.5 Why stochastic diagonal motion repairs the metric

The shortest-path metric induced by four unit-cost cardinal actions is Manhattan distance,

$$d_1[(x,y),(x',y')] = |x-x'| + |y-y'|. \tag{69}$$

This recovers the topology and ordering of a square grid but not its ordinary Euclidean metric

$$d_2[(x,y),(x',y')] = \sqrt{(x-x')^2 + (y-y')^2}. \tag{70}$$

In particular, both an axial step and a diagonal step cost one if all eight moves are deterministic, whereas Euclidean geometry requires the diagonal to cost $\sqrt{2}$ times as much.

The instrument supplies a simple operational unit conversion. Repeating a legal diagonal action with independent success probability p_d gives a geometric waiting time $N \in \{1, 2, \ldots\}$:

$$\mathbb{P}(N = k) = (1-p_d)^{k-1} p_d. \tag{71}$$

Its expectation can be derived from the geometric series,

$$\mathbb{E}N = \sum_{k=1}^{\infty} k(1-p_d)^{k-1} p_d = p_d \frac{d}{dq} \sum_{k=0}^{\infty} q^k \bigg|_{q=1-p_d} \tag{72}$$

$$= p_d \frac{1}{[1-(1-p_d)]^2} = \frac{1}{p_d}. \tag{73}$$

Matching this expected diagonal cost to $\sqrt{2}$ predicts

$$p_d \approx \frac{1}{\sqrt{2}} = 0.707106\ldots. \tag{74}$$

This analytic argument is a sanity check, not the fitted result. Open boundaries and a finite 3×3 set mean that a value minimizing global MDS error need not equal $1/\sqrt{2}$ exactly.

7.6 Exact stochastic shortest paths in the outer loop

For a fixed p_d, the privileged design model constructs a directed weighted graph over the nine localized states. A legal axial edge has expected cost one and a legal diagonal edge has expected cost $1/p_d$. The optimal exact movement cost obeys

$$d^*_{ij} = \min_{a \in \mathcal{A}_i} \left[c(i,a) + d^*_{f_a(i),j} \right], \qquad d^*_{jj} = 0, \tag{75}$$

where $c = 1$ or $1/p_d$. This is the deterministic weighted-graph form obtained after analytically integrating the self-loop failures. Dijkstra's algorithm computes every source–goal pair because costs are nonnegative.

This exact matrix is used only for economical outer search and as a later calibration target. The RL agent is not given the matrix, probabilities, graph, or Dijkstra policy. It samples actual successes and failures and learns from reward. Separating exact design from learned verification prevents a favorable analytic construction from being mistaken for evidence that finite-data behavior recovered it.

7.7 Classical MDS, Euclidean spectra, and stress refinement

Classical MDS gives an algebraic test of Euclidean realizability [24, 25]. Let $\Delta^{(2)}$ contain squared symmetric dissimilarities, let

$$H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top \tag{76}$$

center vectors by subtracting their mean, and define the double-centered Gram matrix

$$B = -\frac{1}{2}H\Delta^{(2)}H. \tag{77}$$

To see why, suppose centered coordinates form the rows of X. Then $\delta_{ij}^2 = x_i^\top x_i + x_j^\top x_j - 2x_i^\top x_j$; double centering cancels the norm terms and leaves $B = XX^\top$. Therefore exact Euclidean distances produce a positive-semidefinite B. If

$$B = V\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n)V^\top, \qquad \lambda_1 \geq \cdots \geq \lambda_n, \tag{78}$$

then the best classical d-dimensional coordinates retain the leading positive eigenvalues:

$$X_d = V_{1:d}\,\mathrm{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_d}). \tag{79}$$

Classical MDS optimizes an algebraic strain criterion, not exactly the raw stress in equation (57). We use X_d as a deterministic initialization for SMACOF (scaling by majorizing a complicated function), an iterative majorization algorithm that monotonically reduces raw stress. Deterministic initialization avoids making reported dimension gaps depend on arbitrary plot seeds.

Negative eigenvalues of B are informative. They cannot be represented as squared norms in a real Euclidean space, so a large negative spectrum indicates that the dissimilarity is non-Euclidean even before choosing dimension. We report

$$V_2^+ = \frac{\lambda_1^+ + \lambda_2^+}{\sum_k \lambda_k^+}, \qquad \lambda^+ = \max(\lambda, 0), \tag{80}$$

$$P_- = \frac{\sum_k |\min(\lambda_k, 0)|}{\sum_k |\lambda_k|}. \tag{81}$$

V_2^+ asks how much representable Euclidean variation lies in two axes; P_- asks how much of the centered relation resists any Euclidean interpretation. They answer different questions and complement stress.

7.8 Validation metrics and why each is present

No single scalar certifies emergent space, so the study reports a diagnostic panel:

All-pairs success.
The fraction of 81 ordered source–goal tasks completed by horizon 12. This tests whether every entry of the cost matrix describes meaningful navigation.

Stress in 1D, 2D, and 3D.
These values test equation (58) and quantify absolute distance distortion at each candidate dimension.

Positive two-dimensional spectral variance.
Equation (80) tests whether two axes dominate the Euclidean part of the centered dissimilarity.

Negative-spectrum fraction.
Equation (81) detects intrinsically non-Euclidean structure that could be hidden by an attractive low-dimensional projection.

Exact-cost correlation.

Pearson correlation between vectorized learned and exact costs tests whether RL recovered the intended ordering and approximately linear scale. It is not a dimensional test: the Manhattan control can score perfectly.

Directionality.

Before MDS, normalized antisymmetry is retained as

$$A(D) = \frac{\|D - D^\top\|_F}{\|D + D^\top\|_F}. \tag{82}$$

This measures finite-sample, policy, or genuine dynamical irreversibility.

Procrustes recovery.

After learning only, the 2D embedding X is aligned to concealed validation coordinates Y by translation, orthogonal transformation, and uniform scale:

$$\min_{c>0, R^\top R = I, b} \|cXR + \mathbf{1}b^\top - Y\|_F^2. \tag{83}$$

If $X_c^\top Y_c = U\Sigma V^\top$, the optimal orthogonal map is $R = UV^\top$ [26]. We report $R_P^2 = 1 - \mathrm{SSE}_{\mathrm{aligned}}/\|Y_c\|_F^2$. This tests coordinate recovery up to the gauge freedoms of distance geometry.

Concealed Euclidean distance correlation.

This compares pairwise distances without choosing an alignment and provides a scale-insensitive check on the intended physical interpretation.

Procrustes coordinates are privileged and never enter policy learning. They are useful because a correct emergent geometry is identifiable only up to translation, rotation, reflection, and scale. Yet Procrustes R_P^2 cannot replace stress: Manhattan distances on a square grid can recover the correct point arrangement after alignment while still distorting metric ratios.

7.9 One-parameter outer search

The outer loop evaluates 81 values of p_d evenly spaced from 0.55 to 0.95. For each value it computes the exact all-pairs matrix, fits 2D coordinates, and combines 2D stress, negative spectrum, and privileged coordinate-recovery penalty. The selected value is

$$p_d^\star = 0.715, \qquad \frac{1}{p_d^\star} = 1.3986, \tag{84}$$

close to the analytic $1/\sqrt{2}$ prediction. The analytic value is plotted as a reference but is not an input target to the numerical objective.

The selected exact matrix has 1D/2D/3D stresses 0.4305/0.0365/0.0360, contains 93.4% of its positive Gram-spectrum variance in two dimensions, and correlates 0.995 with concealed Euclidean distance. The cardinal control has 2D stress 0.1416 and negative-spectrum fraction 0.217, versus 0.081 for the selected instrument. Thus a grid-like arrangement does not by itself make the operational metric Euclidean.

These exact numbers establish that the designed task is suitable for the inner learning experiment. One dimension fails badly: nine points on a line cannot simultaneously preserve the corner, edge, and center relations. Moving from two to three dimensions changes stress by only 0.0005, so the small remaining error is not substantially repaired by an extra spatial axis. The 0.995 distance correlation says the cost ordering and relative scale closely match the concealed Euclidean lattice, while the nonzero negative spectrum records the unavoidable residual of a small weighted graph rather than a continuous plane. The claim is approximate Euclidean compatibility, not exact isometry.

7.10 Inner learning problem and policy extraction

For the place-observed worlds, the learner is the standard tabular Q-learning algorithm of equation (22) applied to state equation (68), with $\alpha = 0.1$, $\gamma = 0.95$, and epsilon-greedy exploration decaying from 1.0 to 0.05. Source and target are sampled independently and uniformly for each of 6,000 episodes. Uniform coverage gives every ordered relation a chance to affect the Q table; it also includes source-equals-goal episodes, which teach the common verification action.

The learned reward policy is

$$\widehat{\pi}_g(a|\ell) = \nVdash\left[a \in \arg\max_{a'} Q(g, J, \ell, a')\right] \tag{85}$$

with deterministic tie-breaking at evaluation. Notice what is and is not learned. The table learns action values and thereby local navigation. MDS does not provide input to the policy and the Q table has no x or y columns. Conversely, the final spatial coordinates are computed from empirical policy costs, not read from the table or hidden basis labels. Control learning and geometric reconstruction are two distinct stages.

For each ordered pair (i, j), evaluation resets to source i, executes $\widehat{\pi}_{g_j}$ without exploration, and repeats 100 times. Let $\widehat{s}_{ij}$ be the success fraction and let $\widetilde{\tau}_{ij}^{(m)} = \min(\tau_{ij}^{(m)}, T)$ assign the horizon to a failure. The empirical restricted cost is

$$\widehat{C}_{ij} = \frac{1}{M}\sum_{m=1}^{M} \widetilde{\tau}_{ij}^{(m)}, \qquad M = 100, quadT = 12. \tag{86}$$

The common final-probe cost is then subtracted as in equation (53). Using the restricted rather than success-conditional mean ensures failure increases distance instead of disappearing from it.

MDS requires a symmetric dissimilarity, whereas finite policy estimates are ordered. We therefore form

$$\widehat{D}_{ij}^{\mathrm{sym}} = \frac{1}{2}(\widehat{D}_{ij} + \widehat{D}_{ji}) \tag{87}$$

for the Euclidean hypothesis and separately report equation (82). Symmetrization is a declared projection onto the class of ordinary metric candidates, not evidence that the original process was reversible.

7.11 Matched controls and experimental scale

The three worlds differ one ingredient at a time:

1. **optimized place-observed**: four axial and four cost-matched diagonal actions; a movement reports its destination symbol;
2. **optimized blind**: identical hidden movement Kraus maps and p_d, but a movement reports only success or failure;
3. **cardinal place-observed**: destination symbols remain available, but diagonal actions are removed, leaving the Manhattan metric.

The blind comparison targets observability and state estimation. The cardinal comparison targets metric isotropy while holding localization and basic competence favorable. Together they distinguish "there is a grid," "the agent can navigate," and "the learned cost is Euclidean-like."

Each of three worlds and three seeds receives 6,000 uniformly sampled source–goal episodes. Evaluation uses 100 trials for all 81 ordered pairs: 54,000 training and 72,900 evaluation episodes in total. Failures contribute a finite restricted-mean penalty at horizon 12. The learned matrix is symmetrized for MDS, while normalized antisymmetry is retained separately.

Table 4: Learned spatial geometry, mean ± sample standard deviation over three seeds. R_P^2 aligns MDS coordinates with concealed coordinates only after training.

Metric	Optimized observed	Optimized blind	Cardinal observed
All-pairs success	$1.000 \pm .000$	$.482 \pm .023$	$1.000 \pm .000$
1D stress	$.372 \pm .011$	$.450 \pm .042$	$.460 \pm .000$
2D stress	$\mathbf{.071 \pm .011}$	$.233 \pm .032$	$.142 \pm .000$
3D stress	$.064 \pm .011$	$.228 \pm .038$	$.141 \pm .000$
Procrustes R_P^2	$.975 \pm .005$	$.965 \pm .011$	$1.000 \pm .000$
Exact-cost correlation	$.936 \pm .003$	$.865 \pm .010$	$1.000 \pm .000$
Directionality	$.122 \pm .037$	$.264 \pm .093$	$.000 \pm .000$

The resulting total is 54,000 training episodes and 72,900 evaluation episodes. Three seeds are still too few for narrow confidence intervals, but exhaustive all-pairs evaluation within each run is much stronger than evaluating only the canonical center reset. Every aggregate in table 4 is the mean and sample standard deviation of three independently trained policies.

7.12 Results: competence before geometry

The proposed place-observed learner achieves all 81 ordered tasks in every seed. This result removes the most serious ambiguity in a hitting-cost matrix: none of its entries is dominated by the failure penalty. The cardinal control also reaches 100%, proving that navigation success alone does not select the proposed metric. By contrast, the blind learner succeeds on only 0.482 ± 0.023 of ordered trials. Its poor geometric fit must therefore be read partly as a control failure and not purely as a change of intrinsic dimension.

Why does observation matter so much? A destination symbol makes equation (68) Markov: after every move the same symbol implies the same future transition law, independent of arrival route. Success/failure alone does not. The blind learner must infer place from a bounded action/outcome history, and different hidden sites can produce identical recent binary records. A policy conditioned on such an aliased state must compromise between incompatible actions. This degrades success, inflates restricted costs, and introduces direction-dependent sampling error.

7.13 Results: selecting two dimensions

The proposed construction has a decisive dimension gap: learned stress falls from 0.372 in 1D to 0.071 in 2D, while a third coordinate improves it only to 0.064. Procrustes $R^2 = 0.975$ confirms recovery of the concealed arrangement up to rotation, reflection, translation, and scale. Coordinate recovery alone would be misleading: the cardinal control has $R_P^2 \simeq 1$ but twice the 2D stress because its metric is Manhattan.

The first stress drop is the main dimensional evidence. In relative terms, two dimensions remove about 81% of the one-dimensional normalized stress, whereas a third dimension removes only about 10% of the remaining stress. The residual 0.071 is expected to exceed the exact value 0.0365: the learned policy sometimes selects a longer route, diagonal travel is sampled rather than analytically averaged, and each ordered pair has finite evaluation data. The small standard deviation across seeds indicates that the dimension pattern is more stable than the orientation of any individual embedding.

Procrustes $R_P^2 = 0.975$ means that after removing geometrically irrelevant translation, orthogonal frame, and scale, only about 2.5% of concealed coordinate sum-of-squares remains as alignment error.

This is strong recovery of point arrangement. The cardinal control's perfect value is the decisive caution: an embedding can put corners, edges, and center in their expected visual locations while systematically misrepresenting how far apart they are. Stress, not visual recognizability, detects that difference.

The learned/exact cost correlation 0.936 ± 0.003 asks whether RL recovered the intended stochastic shortest-path problem. It is high but not perfect, consistent with a near-optimal policy and finite geometric waiting times. The cardinal control reaches correlation one because its deterministic tabular problem is learned exactly. Yet its stress remains 0.142. Calibration to a non-Euclidean target cannot make the target Euclidean; this is why exact-cost correlation and metric fit belong in separate columns.

Directionality is 0.122 ± 0.037 in the proposed world. The underlying exact cost is symmetric, so the learned asymmetry is not interpreted as physical time orientation. It reflects independent finite samples, stochastic diagonals, tie-breaking among near-equivalent routes, and imperfect policies. The deterministic cardinal control has zero directionality. The blind value is larger because state aliasing makes policy errors depend strongly on source and target. Reporting this discarded component makes the symmetry assumption auditable.

Figure 11 should be read relationally. Rotating or reflecting a row changes no pairwise distance, and therefore no operational claim. The proposed embeddings consistently separate corner, edge, and center relations in two axes. The blind embeddings retain a rough lattice because some transition information remains inferable, but variable failures stretch and shear the cost relation. The cardinal embeddings look exceptionally regular while their distance matrices remain Manhattan-like. The side-by-side display is deliberately designed to train visual skepticism: regular pictures are hypotheses whose distortion must be quantified.

7.14 Trajectories as motion in the reconstructed coordinates

Having inferred X from the complete symmetrized cost matrix, we can map a new policy rollout into that coordinate system. If its observed place sequence is $(\ell_0, \ell_1, \ldots, \ell_m)$, the plotted path is

$$(x_{\ell_0}, x_{\ell_1}, \ldots, x_{\ell_m}). \tag{88}$$

This is not a latent-state decoder trained on coordinates. It is a lookup from operational place symbols to MDS coordinates derived solely from learned all-pairs difficulty. Movement failures appear as repeated points; successful local actions connect nearby reconstructed places; the final probe verifies the target without adding movement distance.

The displayed policies make one or two movement interventions followed by the common probe. Their coordinate traces are short local paths toward the green goal. Stochasticity is operationally visible: a diagonal action may leave the agent at the same place before a later success. In this precise and limited sense, an action history has become motion through emergent space. The plot does not imply continuous dynamics between points, a universal external frame, or that the hidden Hilbert basis itself was learned.

7.15 Failed constructions and the corrections they motivated

Negative stages were retained because each rules out an initially tempting but insufficient definition of emergent space.

A single origin produces routes, not a space. The first learner always reset at the central state. It achieved roughly 0.92–0.95 success on that reset distribution, which at first looked encouraging.

When evaluated from arbitrary sources, success fell to about 0.25. The policy had learned a star of source-specific routes from the center and had too little evidence for local behavior elsewhere. More fundamentally, one source determines only $D_{\mathrm{center},j}$. A 2D geometry requires relations among all places. Uniform source–goal training corrected both the coverage defect and the conceptual definition.

A hidden spatial graph is not automatically an agent's space. With uniform sources but binary success/failure movement observations, the learner reached only 0.482 all-pairs success. The same Kraus connectivity that supports the positive result was present, yet it was not operationally available through the learner's finite memory. This failure separates a space in the simulator designer's representation from a hodological space usable by the agent. It motivates recurrent predictive localization as the next, more ambitious experiment.

More remembered detail can make geometry worse. When destination symbols were introduced but the agent retained a literal six-event history, navigation success rose to approximately 0.998. Its 2D stress nevertheless remained near 0.16. The state table assigned distinct keys to the same current place depending on arrival route. Those keys received unequal data and learned slightly different policies and costs, fragmenting one operational place into many representational states. The failure illustrates that memory capacity and state quality are not synonymous.

Quotienting by future affordance repairs the representation. The final PlaceQAgent keeps only (g, J, ℓ). Histories ending at the same place symbol are pooled because they imply the same next-outcome law for every action. This predictive equivalence increases sample sharing, reaches 100% all-pairs success, and lowers mean 2D stress to 0.071. The correction is not coordinate supervision; it is the minimal Markov quotient supported by the observation interface.

Together these stages identify three conditions for the present spatial result: broad all-pairs experience, enough observation or memory for operational localization, and a representation that merges histories with the same future affordances. The tuned diagonal cost is a fourth condition for Euclidean rather than merely grid-topological structure.

7.16 Exactly what this construction establishes

The positive conclusion can now be stated without either underselling or overclaiming it. There exists a valid nine-dimensional quantum instrument family, a common-probe goal repertoire, and a coordinate-free RL state for which independently learned all-pairs intervention cost strongly prefers two dimensions, closely recovers a concealed square arrangement, and maps learned action histories to stochastic local paths. Matched controls show that competence, localization, arrangement recovery, and Euclidean metric quality are distinct.

The result is an existence proof in a deliberately classical corner of quantum theory. Localized basis states, a sharp place outcome, open boundaries, and measure-and-prepare motion were designed. The agent reconstructs their metric relations but does not discover an unanticipated number of places from raw coherent dynamics. A stronger emergence claim will require aliased or weak observations, learned recurrent localization, larger and topologically varied systems, coherent channels, held-out landmarks, and geometry that remains stable when no privileged place basis is built into the simulator.

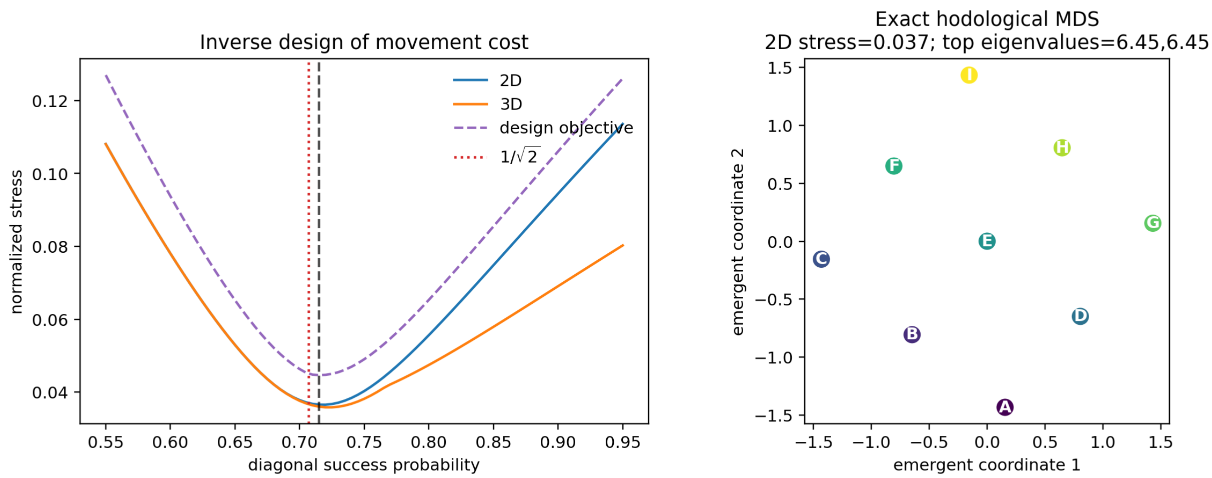


Figure 9: Inverse design of diagonal success probability (left) and exact MDS of the selected hitting-cost matrix (right). The red line is $1/\sqrt{2}$; the black line is the selected 0.715. Exact 2D stress is 0.0365. Place labels carry no coordinate information.

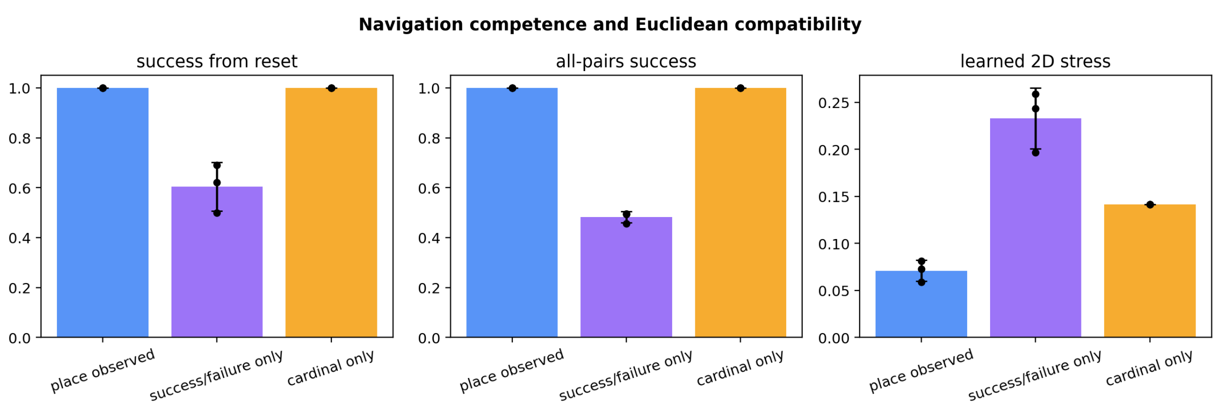


Figure 10: Navigation and Euclidean compatibility. Destination-place observations make the optimized world fully navigable and halve 2D stress relative to the equally competent cardinal/Manhattan control. Binary success/failure observation damages both competence and geometry.

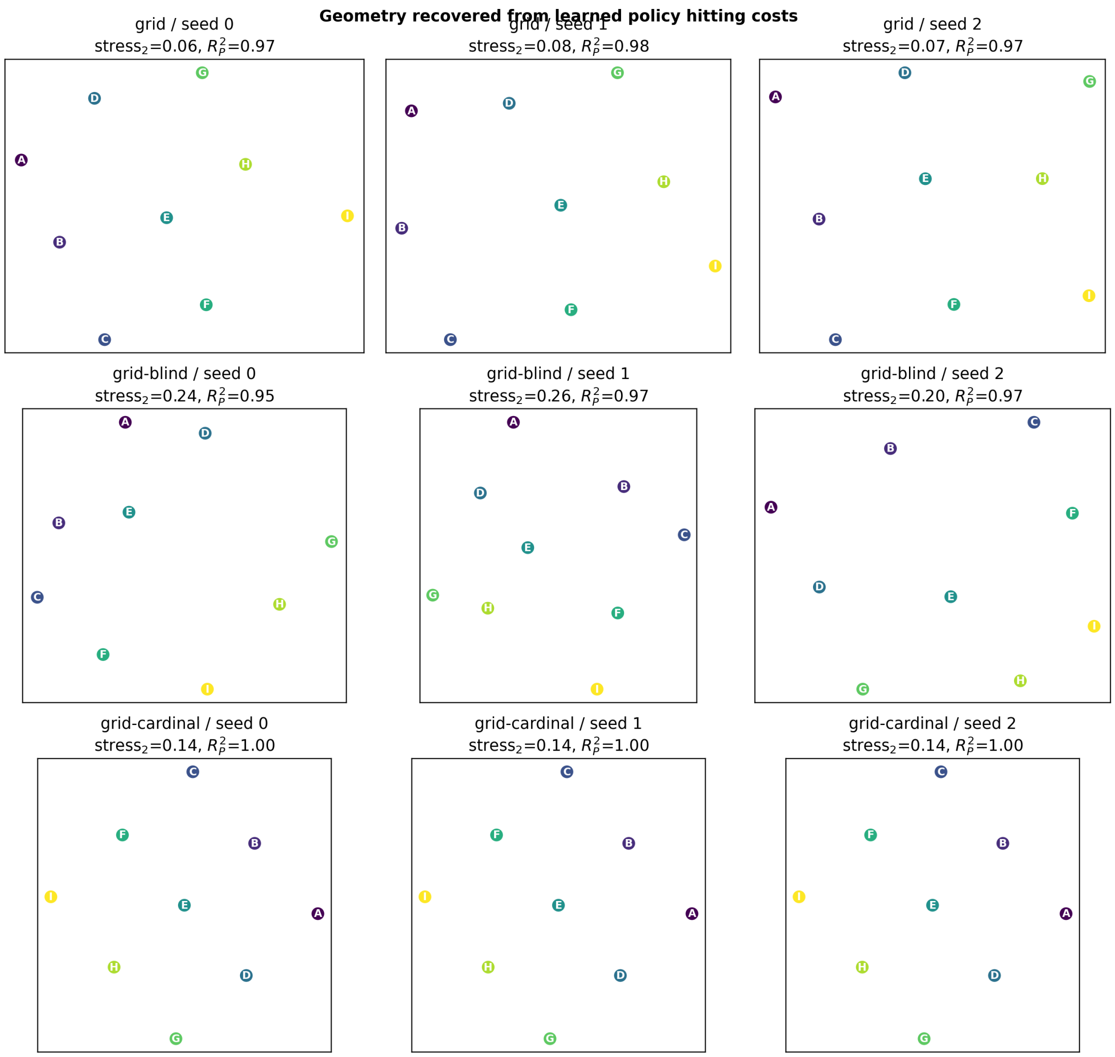


Figure 11: MDS of empirical policy hitting costs. Rows show the proposed world, the partial-observability control, and the cardinal control. Axes have no externally meaningful orientation. The coordinates are outputs of the cost matrix, not inputs to the policies.

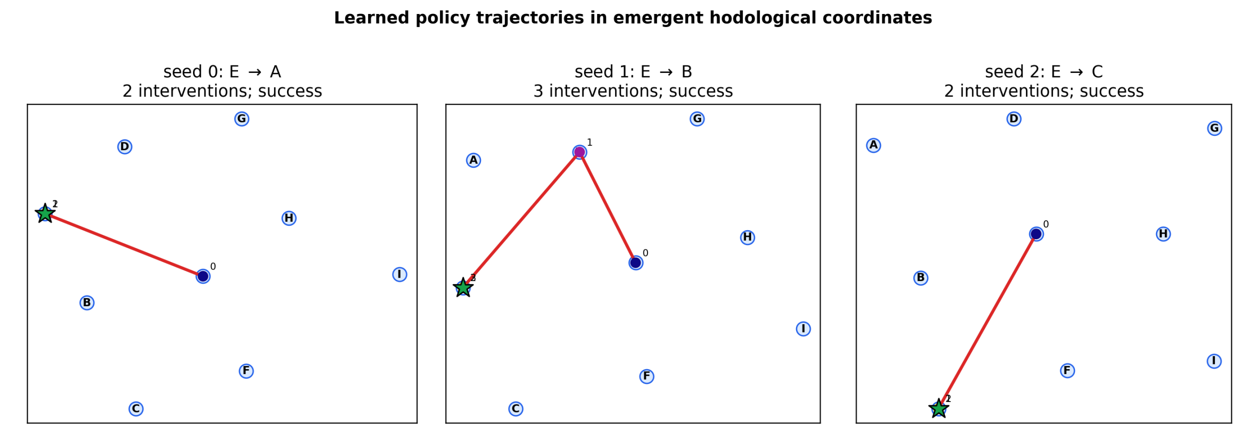


Figure 12: Observed place sequences under learned policies, drawn in coordinates computed only from learned all-pairs difficulty. The green star is the goal; numbers mark intervention order. In this limited sense, action histories are stochastic motion through emergent hodological space. Coordinate frames differ because rotation, reflection, translation, and scale are not operationally fixed.

8 A predictive spatial atlas without online place symbols

8.1 The information-boundary problem

The existence construction in section 7 gives its controller a sharp place symbol before each decision. This is weaker than providing a coordinate: the labels $A, \ldots, I$ carry no adjacency or metric semantics. Nevertheless, the observation already answers the question "where am I?" in a discrete operational sense. If space is to be constructed from an agent's own goal-directed experience, localization should itself be learned.

We therefore retain the same hidden movement geometry and place goals while changing the observation interface. Movement reports only success or failure. Four weak binary probes provide overlapping evidence about place, and a sharp landmark probe is available only as a terminal commitment. The agent must turn the weak history into a predictive belief, propagate that belief through blind motion, and decide when it is ready to claim the goal. This creates a finite partially observable stochastic-shortest-path problem [9]. It also makes the word *place* operational: place is represented by predictions of possible future landmark experiences rather than by a current simulator label.

The online information boundary is

$$\mathcal{I}_t = \{g, (a_1, o_1), \ldots, (a_t, o_t)\}. \tag{89}$$

The controller is not given ho_t, any Kraus operator, response probabilities, lattice coordinates, or the exact landmark that the terminal probe would report. Source preparation and hidden-state decoding are used only to construct balanced experiments and audit results. The terminal report can label a completed calibration scan or score a navigation commitment, but it cannot inform a later action in that episode.

8.2 Weak quantum beacons

Let the concealed site basis again be $\{|s\rangle : s = 0, \ldots, 8\}$. Beacon action $b \in \{1, \ldots, 4\}$ is a binary quantum instrument with diagonal Kraus operators

$$B_0^{(b)} = \sum_{s=0}^{8} \sqrt{1 - q_{bs}}\, |s\rangle\langle s|, \tag{90}$$

$$B_1^{(b)} = \sum_{s=0}^{8} \sqrt{q_{bs}}\, |s\rangle\langle s|. \tag{91}$$

The validity calculation is immediate but worth stating:

$$B_0^{(b)\dagger} B_0^{(b)} + B_1^{(b)\dagger} B_1^{(b)} = \sum_s [(1 - q_{bs}) + q_{bs}] |s\rangle\langle s| = I. \tag{92}$$

For a localized input $\rho_s = |s\rangle\langle s|$, the probability of report one is

$$\mathbb{P}(O = 1 \mid s, b) = \mathrm{Tr}(B_1^{(b)} \rho_s B_1^{(b)\dagger}) = q_{bs}, \tag{93}$$

and the normalized posterior quantum state remains ρ_s for either report. The beacon is consequently quantum nondemolition (QND) on the localized states: it acquires noisy information without moving the system.

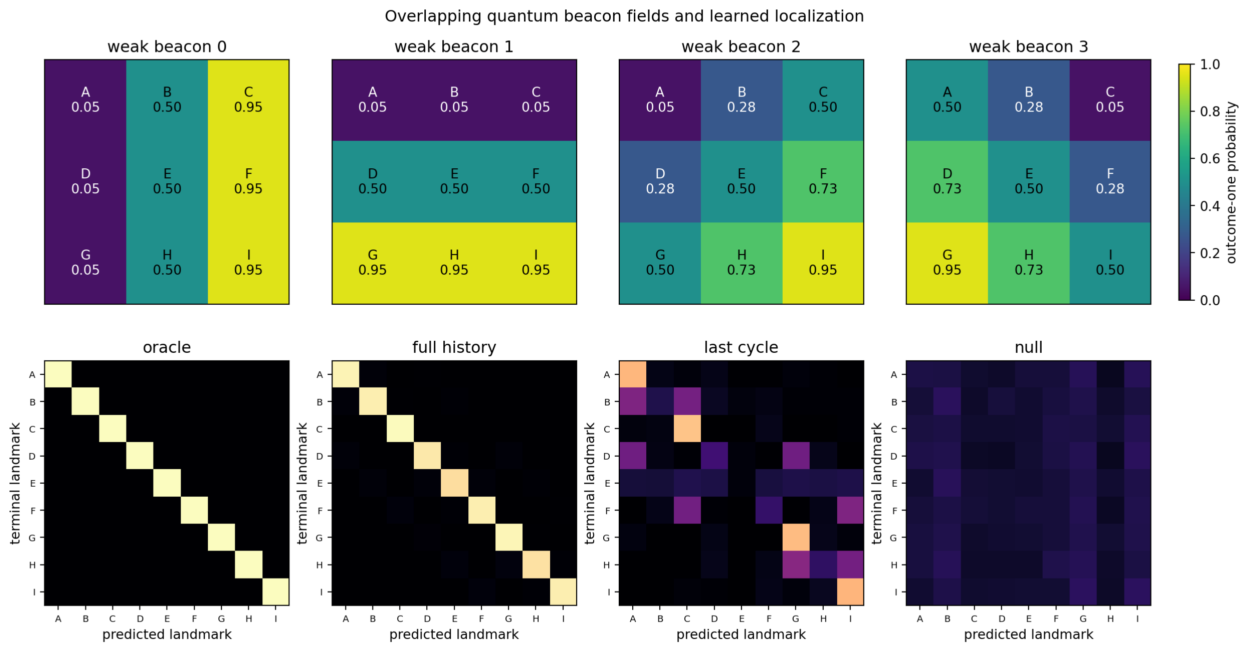


Figure 13: Weak sensing and delayed landmark prediction. Top: the four place-dependent Bernoulli response fields; each sample is ambiguous and the coordinates shown are privileged design labels. Bottom: mean row-normalized held-out confusion matrices. Twelve-cycle history nearly resolves the nine landmarks, one retained cycle preserves broad aliases, and place-independent beacons remain at chance.

The four fields vary horizontally, vertically, diagonally, and anti-diagonally. Their response probabilities lie between 0.05 and 0.95. No single binary report identifies one of nine alternatives; even one complete four-beacon cycle is strongly ambiguous. Yet the response vectors

$$q_s = (q_{1s}, q_{2s}, q_{3s}, q_{4s}) \tag{94}$$

are distinct. Repetition accumulates statistical evidence while leaving the localized state fixed. This is the simplest controlled realization of a weak landmark field.

For comparison, the null environment sets $q_{bs} = 1/2$ for every b, s. Its scan has identical length, alphabet, model capacity, optimization, and delayed labels, but

$$\mathbb{P}(h \mid s) = \mathbb{P}(h) \tag{95}$$

for every scan history h. No classifier can exceed the uniform-prior Bayes accuracy $1/9$ in expectation.

8.3 Delayed landmark prediction as predictive-state learning

A scan applies the action sequence $(1, 2, 3, 4)^{12}$ and records the 48 action/outcome pairs

$$h = ((b_1, o_1), \ldots, (b_{48}, o_{48})). \tag{96}$$

Only after the scan does the common sharp instrument

$$L_s = |s\rangle\langle s|, \qquad s = 0, \ldots, 8, \tag{97}$$

provide the target. The desired output is

$$\widehat{b}_0(s \mid h) \approx \mathbb{P}(\text{terminal landmark } s \mid h). \tag{98}$$

This is not density-matrix regression. It predicts an intervention/outcome counterfactual that the agent can operationally test. In the language of predictive-state representations, histories are equivalent insofar as they support the same future-test predictions [10].

Each observed pair is converted to a learned token vector x_t. A GRU updates its hidden memory according to

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r), \tag{99}$$

$$u_t = \sigma(W_u x_t + U_u h_{t-1} + b_u), \tag{100}$$

$$\widetilde{h}_t = \tanh[W_h x_t + U_h(r_t \odot h_{t-1}) + b_h], \tag{101}$$

$$h_t = (1 - u_t) \odot h_{t-1} + u_t \odot \widetilde{h}_t. \tag{102}$$

The reset gate r_t controls how much previous memory enters the candidate; the update gate u_t interpolates between old and proposed memory. A softmax classifier maps the final state to nine probabilities,

$$\widehat{b}_0(s \mid h) = \frac{\exp(w_s^\top h_{48} + c_s)}{\sum_j \exp(w_j^\top h_{48} + c_j)}. \tag{103}$$

Minimizing negative log likelihood,

$$\mathcal{L}_{\text{loc}} = -\frac{1}{N} \sum_{i=1}^{N} \log \widehat{b}_0(s_i \mid h_i), \tag{104}$$

is proper: in the population limit it is uniquely minimized by the true conditional distribution. Accuracy tests the maximum-probability decision, whereas NLL strongly penalizes confident mistakes. We also report the multiclass Brier score

$$\text{Brier} = \frac{1}{N} \sum_i \sum_s [\widehat{b}_0(s \mid h_i) - \mathbf{1}\{s_i = s\}]^2, \tag{105}$$

posterior confidence, and entropy. These metrics distinguish useful uncertainty estimates from merely correct class labels.

Because the simulator's q_{bs} are known to the offline analyst, an exact Bayes ceiling can be computed by multiplying Bernoulli likelihoods. It is not used for training. Agreement between the learned and Bayes accuracies tells us whether residual error comes from insufficient observations or failed optimization.

8.4 Learning a movement model from landmark-anchored surveys

Localization alone is not navigation. The agent must know how its interventions transform the predictive place distribution. For each previously verified landmark s and blind movement a, a survey records the binary movement outcome o and then terminates with landmark s'. Repeated trials estimate the joint controlled kernel

$$\widehat{T}_a(o, s' \mid s) = \widehat{\mathbb{P}}(O_t = o, S_{t+1} = s' \mid S_t = s, A_t = a). \tag{106}$$

The destination probe is delayed until after the move, so coordinates and Kraus events are unnecessary. However, the verified source and destination landmarks are supervision. This phase is best understood as empirical chart calibration, not autonomous discovery of the chart's labels.

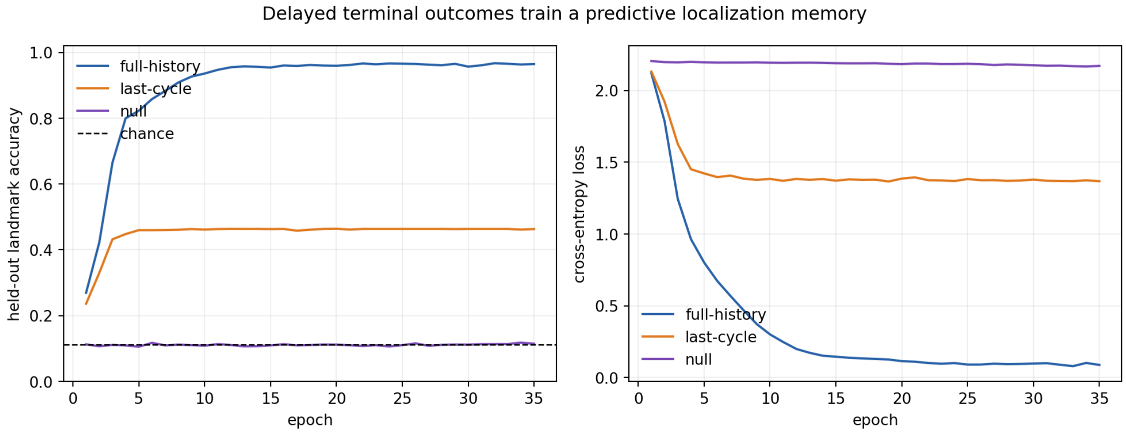


Figure 14: Training and held-out delayed-landmark performance across three seeds. Full-history prediction rapidly exploits accumulated evidence. The last-cycle and null controls plateau near their respective Bayes limits, so their poor localization is not plausibly explained by undertraining.

Marginalizing the report yields the planning transition model,

$$\widehat{P}_a(s' \mid s) = \sum_o \widehat{T}_a(o, s' \mid s). \tag{107}$$

Retaining the joint tensor remains essential at execution time. Given belief $b_t(s)$, selected movement a_t, and observed report o_t, Bayes filtering gives

$$b_{t+1}(s') = \frac{\sum_s b_t(s)\widehat{T}_{a_t}(o_t, s' \mid s)}{\sum_{j,s} b_t(s)\widehat{T}_{a_t}(o_t, j \mid s)}. \tag{108}$$

Thus a success report and a failure report generally produce different predictive positions even though neither names a destination. The mean total-variation discrepancy between the empirical and privileged exact kernels is only 0.0144 ± 0.0008, indicating that transition-estimation error is small in the production regime.

8.5 Belief-state stochastic-shortest-path planning

For every goal landmark g, value iteration solves

$$V_g(g) = 0, \tag{109}$$

$$V_g(s) = \min_a \left[1 + \sum_{s'} \widehat{P}_a(s' \mid s)V_g(s')\right], \qquad s \neq g, \tag{110}$$

with action cost

$$Q_g(s, a) = 1 + \sum_{s'} \widehat{P}_a(s' \mid s)V_g(s'). \tag{111}$$

These are stochastic-shortest-path rather than discounted-reward equations. One unit is charged for every movement; the absorbing goal has zero future movement cost. The online action rule averages over uncertainty,

$$a_t^* = \arg\min_a \sum_s b_t(s)Q_g(s, a). \tag{112}$$

When the posterior mode equals g, the controller invokes the terminal probe. A correct outcome succeeds and an incorrect outcome fails immediately.

Equation (112) is a tractable approximation to exact belief-MDP planning. Exact planning would assign a value directly to every continuous belief and account for the future informational value of movement outcomes. Here state-level costs are computed first and linearly averaged. The approximation is appropriate when initial beliefs are sharp and dynamics are nearly deterministic, but it is not generally optimal. Its empirical success is therefore evaluated rather than assumed.

The controller performs no further beacon scans after its initial scan. New uncertainty enters only through blind movements and is propagated by equation (108). This makes saved paths sequences of predictive beliefs, not privileged state labels.

8.6 Experimental design and matched controls

The production study uses three fixed seeds. For each neural condition and seed, 400 calibration and 200 held-out scans are generated per landmark and the GRU is trained for 35 epochs. Each scan contains twelve cycles. Movement surveys use 100 trials for every one of nine sources and eight actions. Frozen navigation uses 100 episodes for each of 81 ordered source–goal pairs with at most twelve movements.

The resulting scale is 21,600 transition-survey trials and 97,200 navigation episodes. Navigation alone contains 4,665,600 weak-beacon outcomes. Nine model checkpoints, all per-seed confusion/cost/success/transition matrices, and complete trajectory audits are retained.

Four navigation conditions isolate distinct questions:

Online oracle
: Starts from a one-hot exact-place belief but uses the same learned transition model and planner. It estimates the loss due specifically to predictive localization.

Full history
: Uses all 48 weak outcomes. This is the proposed atlas.

Last cycle
: Collects the same 48 outcomes but discards the first 44 before inference. Architecture and sensor cost are matched while usable temporal evidence is ablated.

Null fields
: Uses all 48 outcomes from place-independent beacons. This tests whether recurrence, labels, or grid symmetry can produce the result in the absence of localization information.

The movement-cost matrix excludes the fixed 48-probe scan and one terminal commitment because those add common overhead to every pair and would compress relative spatial variation. Total interventions are nevertheless reported: the full-history agent uses approximately 50.5 interventions per episode while its mean movement cost is 1.74. Removing common overhead from geometry is not a claim that sensing is free; it exposes a major efficiency limitation.

8.7 Localization results

The full-history classifier reaches 96.4% held-out accuracy, within 1.3 percentage points of the exact Bayes classifier. Its mean posterior confidence is 0.961 ± 0.003, close to its accuracy, and its mean entropy is 0.144 ± 0.008 nats. The recurrent representation therefore extracts most of the place information actually present in the finite scan and is not simply memorizing calibration examples.

The last-cycle accuracy of 46.3% equals its Bayes ceiling. This is especially diagnostic: the model has learned as much as it can from four retained samples, but one cycle does not sufficiently

Table 5: Delayed landmark prediction, mean ± sample standard deviation over three seeds. Bayes accuracy uses privileged beacon likelihoods only as an offline information ceiling. Lower NLL and Brier score are better.

Condition	cycles used	accuracy	Bayes accuracy	NLL	Brier
Full history	12	$\mathbf{0.964 \pm 0.008}$	0.977 ± 0.004	0.112 ± 0.017	0.055 ± 0.010
Last cycle	1	0.463 ± 0.006	0.463 ± 0.006	1.386 ± 0.025	0.685 ± 0.006
Null fields	12	0.114 ± 0.008	0.111	2.229 ± 0.005	0.896 ± 0.001

Table 6: Frozen all-pairs navigation and geometry, mean ± sample standard deviation across three seeds. Cost correlations and Procrustes scores use privileged exact quantities only after learning. Geometry is interpreted only after success.

Condition	success	move cost	exact-cost r	stress 1D	stress 2D	stress 3D	Procrustes R^2
Oracle	$1.000 \pm .001$	$1.485 \pm .006$	$0.998 \pm .001$	$0.418 \pm .012$	$0.040 \pm .006$	$0.039 \pm .005$	$1.000 \pm .000$
Full history	$\mathbf{0.973 \pm .004}$	$1.742 \pm .033$	$\mathbf{0.948 \pm .013}$	$0.407 \pm .021$	$\mathbf{0.075 \pm .005}$	$0.043 \pm .004$	$\mathbf{0.987 \pm .005}$
Last cycle	$0.614 \pm .005$	$5.223 \pm .021$	$-0.054 \pm .020$	$0.412 \pm .013$	$0.192 \pm .013$	$0.112 \pm .008$	$0.451 \pm .014$
Null fields	$0.484 \pm .014$	$6.887 \pm .133$	$0.628 \pm .079$	$0.456 \pm .006$	$0.214 \pm .009$	$0.102 \pm .009$	$0.856 \pm .108$

distinguish the landmarks. The null model obtains 11.4%, statistically consistent with the 1/9 ceiling. Long memory is useful only when the history contains place-dependent evidence.

8.8 Navigation, dimension, and geometric fidelity

The full-history atlas succeeds on $97.3 \pm 0.4\%$ of held-out ordered pairs, closing most of the gap to the oracle. Its movement cost is correspondingly only 0.26 actions above the oracle average. More importantly, the learned cost matrix correlates 0.948 ± 0.013 with the exact stochastic-shortest-path matrix. The agent has not merely learned to stop near likely goals; its relative goal difficulties closely recover the designed intervention structure.

The dimension test shows a large stress reduction from one dimension (0.407) to two (0.075). Two coordinates capture 0.870 ± 0.017 of the positive classical-MDS spectrum, concealed Euclidean distance correlation is 0.942 ± 0.012, and coordinate alignment reaches Procrustes $R^2 = 0.987 \pm 0.005$. These independent diagnostics support the interpretation of a recovered two-dimensional spatial base.

The further reduction from 2D stress 0.075 to 3D stress 0.043 is not negligible. Unlike the near-perfect oracle plateau, predictive localization errors induce extra nonspatial distortion. The proper statement is therefore that the atlas strongly recovers a designed 2D organization, not that its finite-sample dissimilarity matrix has exact Euclidean rank two. This residual provides a quantitative target for better filters and active sensing.

The last-cycle condition reaches only 61.4% success and its cost correlation with the exact map is slightly negative. Its coordinate recovery falls to $R^2 = 0.451$. The null condition reaches 48.4% success with a mean movement cost of 6.89. Its apparently moderate exact-cost correlation and occasionally high Procrustes score are cautionary rather than positive: boundary effects, censoring, and the symmetry of a nine-point grid can imprint structure on an incompetent policy. Its 2D stress is high, success is low, and Procrustes variance is large. A recognizable picture is never accepted without competence, metric fit, and controls.

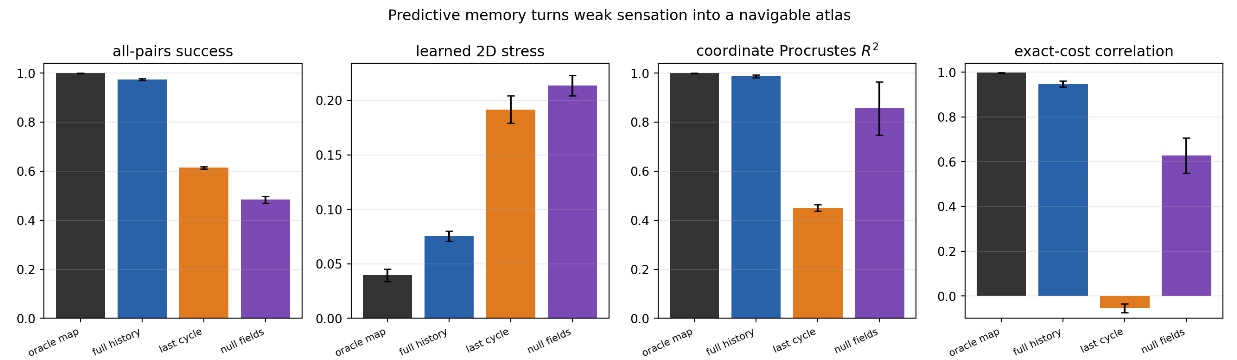


Figure 15: Competence and geometric diagnostics for the oracle, predictive atlas, memory-limited control, and null-sensor control. Error bars show one sample standard deviation across three seeds. The full history nearly matches the oracle; the controls separate accumulated evidence from architecture and computation.

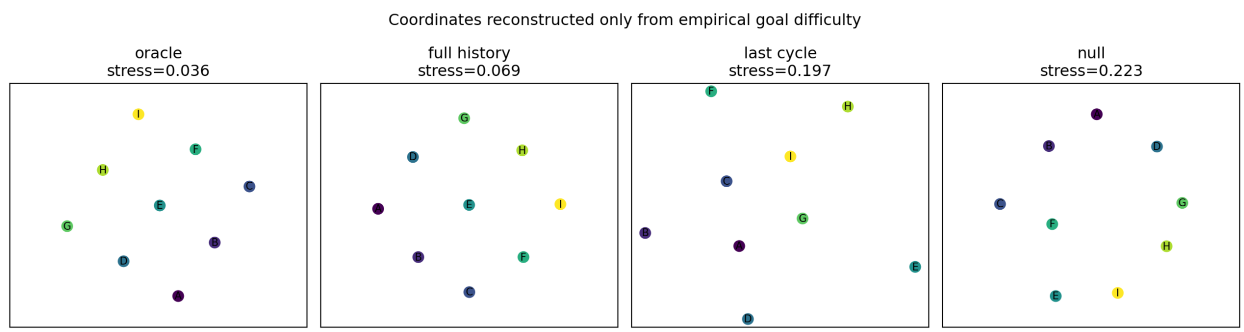


Figure 16: Two-dimensional MDS reconstructions from empirical movement costs. Point locations are fitted from goal difficulty; concealed coordinates are used only to align orientation for evaluation. The full-history atlas retains the lattice organization, while the equal-budget memory control is badly warped.

8.9 Trajectories as motion through a predictive atlas

An MDS plot of pairwise goals does not by itself show that an individual agent moves through the space. For representative center-to-corner tasks we save, after every blind action, the full belief b_t, its maximum-probability landmark, the movement report, and the concealed simulator site for offline audit. We then locate the belief mode in coordinates inferred solely from the full-history cost matrix.

The result makes "trajectory" explicitly agent-relative. The physically privileged trace exists inside the simulator; the operational trajectory is a sequence of changing expectations about future landmark reports. Agreement between them validates the atlas. Disagreement remains meaningful uncertainty, not an undefined agent position.

8.10 The failed pilot and why it matters

The first pilot used eight scan cycles, 200 calibration examples per landmark, and 20 epochs. It obtained 86.5% localization and 88.7% navigation, but 2D stress remained 0.184. Moderate competence did not preserve the metric. This failure motivated a longer scan and independently confirms that success and geometric fidelity are different tests.

A second pilot used twelve cycles, 300 examples per landmark, and 30 epochs. It reached 97.1% localization, 97.7% navigation, 2D stress 0.079, and Procrustes $R^2 = 0.984$. Only then was the

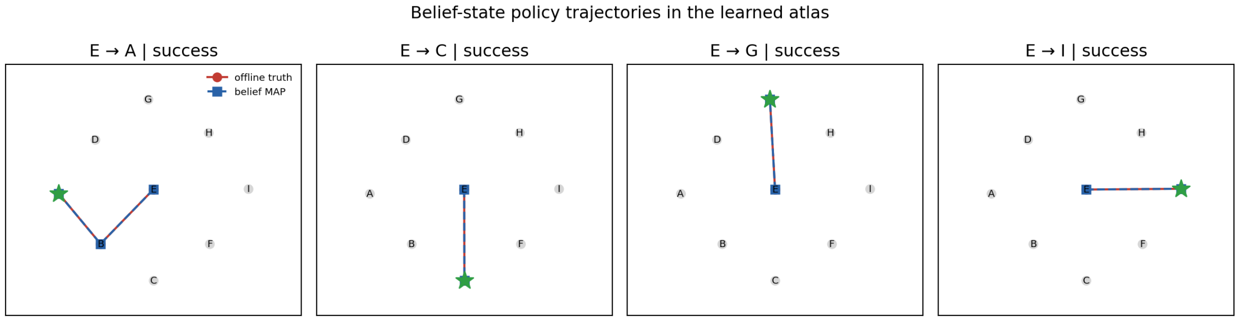


Figure 17: Representative belief-state trajectories. Arrows connect successive maximum-posterior landmarks in the learned goal-cost embedding. The offline true-site trace audits localization but is never supplied to the controller. For these examples, blind outcome filtering produces coherent center-to-corner motion through the predictive atlas.

three-seed production protocol frozen, with 400 calibration examples and 35 epochs. Pilot outputs are not mixed into the production aggregate; their diagnostic values are retained in the project log.

8.11 What the predictive atlas establishes

The predictive atlas strengthens the earlier existence proof. There exists a valid weak-instrument interface for which an agent can learn an operational place posterior from temporal evidence, propagate it through blind stochastic actions, achieve arbitrary landmark goals, and recover an approximately two-dimensional Euclidean goal-cost geometry without receiving an exact online place report. The memory and null controls identify accumulated informative evidence as a necessary causal ingredient.

Several primitives remain designed. The simulator contains nine localized basis states; the terminal outcome supplies supervised landmark identity; transition surveys begin from verified landmarks; beacon instruments commute with the place basis; and movement is entanglement breaking. Hence the agent learns to localize within a supplied landmark vocabulary rather than discovering an unforeseen decomposition of Hilbert space. The fixed scan also costs 48 interventions, far more than the typical journey. The result should be read as the first learned predictive atlas, not as generic emergence of physical space from arbitrary quantum dynamics.

9 Discussion and limitations

Useful measurement strategies can be learned without exposing hidden quantum variables. Recurrent memory sometimes helps, but a short literal history is a strong small-world baseline. Goal embeddings carry moderate behavioral meaning, although local policy geometry transfers more reliably than whole-trajectory geometry. Directed coordinates expose coherent measurement displacements even though scalar cost is poorly calibrated.

The spatial study gives a stronger but narrower positive statement: there exists a valid quantum instrument/goal construction in which policy-derived difficulty selects two dimensions, recovers a concealed lattice, and turns observed action sequences into spatial paths. Its controls show that hidden connectivity, coordinate recovery, control success, and Euclidean metric fit are distinct requirements.

The predictive-atlas successor weakens the observation assumption substantially. Exact online place is replaced by anticipated landmark experience inferred from weak history. Near-oracle navigation and geometric recovery show that a supplied current-state symbol is not necessary on this world. The residual 2D-to-3D stress improvement is equally informative: representation error appears as nonspatial metric distortion, making emergent dimension a joint property of dynamics, sensing, memory, and control.

These results do not show that a Bloch sphere, Hilbert space, or uniquely quantum geometry spontaneously emerged. Such a claim would require latent-state analysis, intervention ablations, classical controls, and privileged offline decoding of quantum invariants. Other limitations are:

- two seeds cannot support significance claims; modern RL methodology recommends more runs, intervals, robust aggregates, and performance profiles [18];
- episode budgets are equal, not compute budgets, and hyperparameters are not independently optimized;
- horizon 15 censors hard goals, reward shaping affects behavior, and conditional steps omit failures;
- one-step prediction does not guarantee predictive sufficiency;
- goal embeddings are lookup entries and cannot represent unseen goals;
- embedding targets come from the same Q-head, creating shared-error and circularity risks despite held-out evaluation;
- cost is learned off-policy from reward-oriented behavior and is not constrained to satisfy stochastic-shortest-path consistency;
- simulators omit Hamiltonian dynamics, decoherence, detector error, latency, and hardware drift.
- the spatial construction uses localized qudit states and entanglement-breaking measure-and-prepare channels; latent places, open boundaries, and a sharp localization probe are designed into the simulator;
- the 3×3 lattice is too small to establish continuum structure, topology recovery, scale dependence, or curvature, and three spatial seeds remain exploratory;
- symmetrizing learned hitting costs is appropriate for testing a Euclidean hypothesis but discards directed information that a more general hodological geometry must retain.
- the predictive atlas uses delayed landmark supervision and landmark-anchored transition surveys; it does not discover its place vocabulary without labels;
- its diagonal QND beacons commute with the latent place basis and its movement channels remain entanglement breaking, so no specifically coherent quantum resource is responsible for localization;
- every navigation episode pays for a fixed 48-probe scan, which is omitted from relative movement geometry but dominates the reported total intervention count;
- belief-weighted state costs approximate rather than solve the full belief-MDP, and the controller does not actively choose additional sensing.

10 Outlook

Compositional goals and transfer. Encode checkpoint sequences with a GRU or transformer, hold out complete goals, and test zero/few-shot transfer. Geometry matters if proximity predicts adaptation speed. Successor features provide a principled comparison [15].

Multi-step predictive state. Train action-conditional predictions over entire future tests, compare recurrent and transformer memory, and measure predictive calibration by horizon. Vary informational completeness to determine which latent dimensions disappear.

Planning in possibility space. Use equation (44) to compare direct Q-control, greedy distance reduction, and tree search through imagined outcomes. The learned geometry should improve decisions, not only visualization.

Distributional reachability. Learn τ_g distributions or $R_g(z, T)$ directly, using proper scoring rules and reliability plots. Distributional Bellman methods are a starting point [16].

Directed non-Euclidean structure. Explore directed graphs, asymmetric energy models, quasimetrics, hyperbolic goal hierarchies, and Finsler-like local costs. Choose geometry through predictive validity and planning utility, not aesthetics.

Optimize states, instruments, and landmarks. Generalize equation (56) from one success probability to positive Choi matrices or Stinespring isometries, enforcing complete positivity and trace preservation. Jointly select goal landmarks for coverage and stable trilateration, penalizing nonlocal teleportation and goal-specific actions. Evolutionary search is a robust first outer loop; differentiable meta-learning becomes attractive once the inner planner is model-based.

From predictive to active localization. Explicit online place localization has now been removed in section 8. The next controller should jointly choose sensing, movement, and commitment under a common cost. Information gain $H(b) - \sum_o P(o \mid b, a) H(b'_{a,o})$ supplies a useful diagnostic, but sensing should ultimately be valued by how much it lowers goal-conditioned cost rather than by entropy alone. The agent should learn its observation and transition models from uninterrupted experience, eliminating separated landmark-anchored surveys. Scaling from grids to tori, cylinders, spheres, and defected graphs should then be evaluated with out-of-sample stress, local dimension, neighborhood preservation, and topology, not only global MDS [27].

Spatial base and internal fiber. Give each place several internal quantum states and goals, then test whether the full hodological state space E locally decomposes as $F_b \hookrightarrow E \xrightarrow{\pi} B$, with a stable 2D or 3D spatial base B and internal fiber F_b [28]. Actions may be horizontal, vertical, or coupled. Learned transition maps between local charts and the failure of an internal state to return after transport around a closed base loop provide operational notions of connection and holonomy.

Curvature and spacetime. Position-dependent transition kernels permit local metric and transport curvature tests; Ollivier curvature supplies one principled discrete diagnostic for Markov processes [29]. This is only a bridge. General relativity relates spacetime curvature to dynamics [30], so a credible analogue would require robust local geometry, connection, sources, and an independently tested field relation. Time cannot simply be appended as a Euclidean coordinate: future-goal reachability must first yield causal order, directed cones, clock tasks, and observer-dependent simultaneity.

Information-seeking action. Replace undirected epsilon exploration with expected information gain, prediction uncertainty reduction, or reachability uncertainty reduction. This turns experimentation itself into goal-directed behavior.

Privileged offline physics. Pair learned z_t with hidden Bloch coordinates only after training. Test decodability, intrinsic dimension, convexity, and equivalence classes; compare against classical partially observed environments.

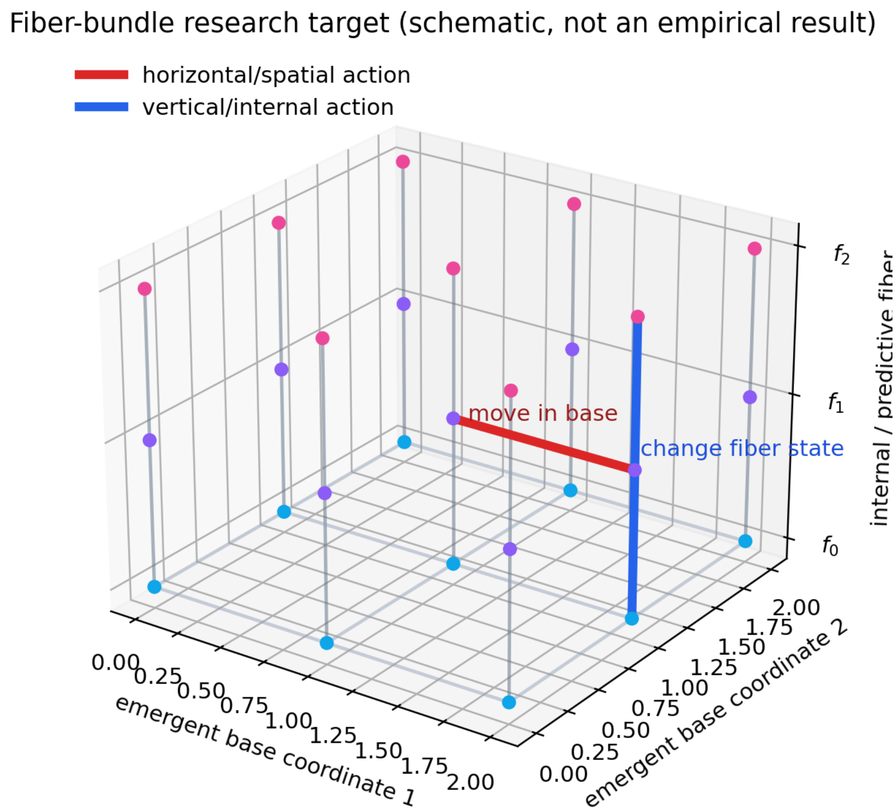


Figure 18: Proposed spatial-base/internal-fiber program. This is a schematic, not an empirical result. Horizontal actions move through emergent place; vertical actions alter internal possibilities.

Stronger statistics and hardware. Use 10–20 seeds, longer budgets, bootstrap intervals, interquartile means, and performance profiles [17, 18]. Then add decoherence, readout error, continuous weak measurement, and device drift, ultimately learning from laboratory records as envisioned in model-free quantum control [20].

11 Conclusion

An agent beginning only with interventions and outcomes can learn more than a policy. It can compress history, predict contemplated measurements, compare aims by behavior, estimate goal-relative reachability, and represent measurement backaction as movement through future possibility. The present study makes that program concrete but also establishes a demanding standard: an attractive embedding is not self-validating. It must predict independent behavior, transfer, and reachability.

The inverse-designed qudit world additionally establishes the first spatial existence result: all-pairs learned difficulty robustly prefers two dimensions and learned policies trace paths in that emergent map. Just as importantly, the failures show that a single-origin task, a hidden grid, or a visually grid-like embedding is not enough. Broad source coverage, operational localization or memory, and approximately isotropic movement costs are jointly required.

The predictive atlas advances that result from reported place to inferred place. Weak, individually ambiguous quantum beacons are integrated into a distribution over future landmark experiences;

blind movement reports update that distribution; and belief-state policies recover a two-dimensional goal-cost map with 97.3% all-pairs success. The short-memory and null-sensor controls show that the map depends on accumulated operational evidence rather than network capacity or a hidden grid alone. Space in this construction is therefore closer to a learned organization of action and anticipation, though delayed labels and a designed place basis remain.

The geometry-aware agent is competitive but not universally superior. Its embedding has measurable, seed-dependent strategic meaning; its directed cost orders some goals but is not accurate in absolute units. These shortcomings identify the next experiments. The long-term prospect is an agent not handed a physical state space, but able to construct a predictive present and a geometry of possible aims from action and consequence alone.

A Reproduction and artifacts

```
conda run -n qbist_spacetime python run_experiment_suite.py \
  --scenarios qubit-zx-weak:one,qubit-zx-weak:plus,\
qubit-pauli:plus-i,qubit-unsharp:mixed,\
qubit-pauli-sic:mixed,qutrit-mub:two \
  --backends tabular,gru,multi-gru --seeds 0,1 \
  --episodes 400 --eval-episodes 100 --geometry-episodes 40 \
  --max-steps 15 --epsilon-decay 0.99 --device cpu \
  --output results/comparative
python plot_results.py results/comparative

python spatial_hodology.py --output results/spatial-hodology \
  --seeds 0,1,2 --episodes 6000 --pair-episodes 100 \
  --max-steps 12

conda run --no-capture-output -n qbist_spacetime \
  python predictive_atlas.py --output results/predictive-atlas \
  --seeds 0,1,2 --scan-cycles 12 \
  --calibration-per-site 400 --test-per-site 200 --epochs 35 \
  --transition-trials 100 --pair-episodes 100 \
  --max-moves 12 --device cpu
python plot_predictive_atlas.py results/predictive-atlas
```

The manifest records exact runs; episode CSVs retain raw outcomes; geometry directories store embeddings, three distance views, reachability, curves, and displacements; and the plot tree contains 51 figures. The spatial bundle separately retains its 81-point instrument search, exact and learned all-pairs matrices, text Q-tables, trajectory records, manifest, and five figures. The predictive-atlas bundle adds three localization summaries/curves, per-seed confusion and transition matrices, four sets of navigation matrices, nine GRU checkpoints, belief trajectories, a manifest that records the online information boundary, and five figures.

B Goal repertoires

qubit-zx-weak

$Z0, Z1, X0, X1, Z0\_X0, X0\_Z0, weakZ0\_Z0.$

qubit-pauli
$Z0, X0, Y0, Z0\_X0, X0\_Z0, Y0\_X1, Z0\_X0\_Y0$.
qubit-unsharp
$wZ0, wX0, wY0, wZ0\_wX0, wX0\_wZ0, wZ0\_wX0\_wY0$.
qubit-pauli-sic
$Z0, X0, Y0, SIC0, SIC1, SIC2, SIC3, Z0\_SIC0, SIC0\_X0$.
qutrit-mub
$Z0, Z1, Z2, F0, F1, Z0\_F0, F0\_Z0, Z0\_F0\_P0$.
spatial worlds
$A, B, C, D, E, F, G, H, I$, each denoting one outcome of the same common place probe.
predictive-atlas worlds
The same nine terminal landmark goals, reached using eight blind movements and inferred from four weak binary beacon fields.

C Worked examples for the new reader

C.1 A numerical Q-learning update

Suppose the current estimate for an observed state–action pair is $Q(s, a) = 2.0$. The action gives reward $r = 1$, the next state is nonterminal, the largest next-state action value is 4, and $\gamma = 0.95$. The TD target and error are

$$Y^Q = 1 + 0.95(4) = 4.8, \qquad \delta^Q = 4.8 - 2.0 = 2.8. \tag{113}$$

With $\alpha = 0.1$, the updated entry is

$$Q_{\text{new}}(s, a) = 2.0 + 0.1(2.8) = 2.28. \tag{114}$$

The target is not treated as certain: one sample moves the estimate only ten percent of the way. Repeated transitions average stochastic outcomes. If the transition terminated, the next-state term would be removed and the target would be just the observed reward.

C.2 Reward value versus intervention cost

Consider two policies for the same goal. Policy A succeeds in two steps with probability one. Policy B succeeds in one step with probability 0.6 and in ten steps otherwise. With only a large completion reward and discounting, policy preference depends on γ and reward design. Their expected undiscounted hitting costs are unambiguous:

$$\mathbb{E}\tau_A = 2, \qquad \mathbb{E}\tau_B = 0.6(1) + 0.4(10) = 4.6. \tag{115}$$

At deadline $T = 1$, however, B has success probability 0.6 while A has zero. This example explains why expected cost, discounted reward, and finite-time success are related but distinct descriptions of goal difficulty.

C.3 Why a square is intrinsically two-dimensional

Take four ideal points at $(0, 0), (1, 0), (0, 1), (1, 1)$. Their Euclidean distance matrix is

$$D = \begin{pmatrix} 0 & 1 & 1 & \sqrt{2} \\ 1 & 0 & \sqrt{2} & 1 \\ 1 & \sqrt{2} & 0 & 1 \\ \sqrt{2} & 1 & 1 & 0 \end{pmatrix}. \tag{116}$$

Double centering $D^{\circ 2}$ using equation (77) yields a positive-semidefinite Gram matrix of rank two. Consequently classical MDS has two positive eigenvalues and reconstructs the square exactly in $\mathbb{R}^2$; a one-dimensional reconstruction must distort at least one pair. If the diagonal entries were changed from $\sqrt{2}$ to 1 while axial entries remained 1, the relation would no longer describe the ordinary unit square even though a plotting algorithm might still arrange the four labels in a square-like pattern. This miniature example is the logic behind the diagonal-cost design and the separation of Procrustes recovery from stress.

C.4 Spatial experiment as an auditable algorithm

The complete study can be summarized without implementation-specific detail:

1. For every candidate p_d, construct the valid movement instruments, compute exact all-pairs stochastic shortest paths, and score their 2D Euclidean compatibility.
2. Fix the best candidate before training. For each random seed and each control world, initialize an empty Q table.
3. Repeatedly sample a source and goal uniformly, emit the coordinate-free source symbol, and train by epsilon-greedy Q-learning for at most 12 actions.
4. Freeze the policy. For each of 81 ordered source–goal pairs, run 100 greedy trials and record success and restricted hitting cost.
5. Subtract the common terminal-probe overhead, retain ordered asymmetry, and symmetrize costs only for the Euclidean test.
6. Fit 1D, 2D, and 3D MDS coordinates; calculate stress, spectrum, calibration, directionality, and privileged Procrustes diagnostics.
7. Map fresh observed policy trajectories into coordinates inferred from the all-pairs costs. Never use concealed coordinates to select an action.

This ordering prevents test information from entering the policy and makes each transformation from quantum instrument to geometric claim inspectable.

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